Monetary Policy Surprises with Imperfect Information

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Abstract

High-frequency interest rate surprises around Fed announcements need not reflect pure monetary policy (MP) shocks under imperfect information: they may also contain central bank information (CBI) shocks stemming from differences between market and Fed assessments of the economy, and Fed-response-to-news (FRN) shocks arising from market misperception of the policy rule. We show that CBI and FRN are not alternatives, as they yield distinct empirical signatures. Using a broad set of high-frequency surprises around FOMC and non-FOMC communications, we jointly identify all three shocks by exploiting (i) the co-movement of rate and equity surprises, (ii) the predictability of surprises from public news, and (iii) heteroskedasticity across announcement types. Dynamic effects are estimated with daily local projections and a monthly Bayesian VAR. The data reveal significant CBI effects at both daily and monthly frequencies, while FRN is relevant only at daily frequency. The resulting monetary policy shock, purged of CBI and FRN effects, produces impulse responses consistent with theory.

JEL codes: E31, E32, E52

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1. Introduction

Assessing the macroeconomic impact of identified monetary policy (MP) shocks is crucial for understanding policy transmission and distinguishing between competing macroeconomic models (Christiano et al. 2005; Gagliardone and Gertler 2023). An informative and popular proxy of such shocks are high-frequency interest-rate surprises around the announcements of the Federal Reserve (Fed) (Kuttner 2001; Gürkaynak et al. 2005b; Gertler and Karadi 2015). However, this proxy can be contaminated with contemporaneously changing market perception about the economic outlook due to Fed communication, a view that is often referred to as the central bank information effect (CBI, Romer and Romer 2000; Jarociński and Karadi 2020). And a further source of contamination can come from market misperception of the systematic policy rule (Miranda-Agrippino and Ricco 2021; Hack et al. 2024). The latter channel has recently been forcefully advocated as a cause of the 'Fed response to news (FRN)' effect by Bauer and Swanson (2023a,b). This view argues that the Fed's unexpectedly aggressive systematic policy response coupled with omitted recent public economic news can be mismeasured as central bank information effect, even though Fed news is actually only about policy. In this paper, we build a unified framework that encompasses MP, CBI and FRN shocks, propose a new approach to simultaneously identify them and we contrast their contributions to the economy.

Our analysis confirms the robust presence of the central bank information shocks, even when the FRN effect is carefully controlled for. Furthermore, even though FRN shocks play a significant role in some specifications – in daily local projections, and monthly VARs using surprises around FOMC events – their impact in our baseline monthly VAR with both FOMC and non-FOMC events is marginal. This can be the result of the stimulative impact of favorable fresh public news and the adverse impact of tighter policy rule offsetting each other. Monetary policy shocks cleaned from the impact of both CBI and FRN shocks generate a downturn in activity, a reduction in the price level and a deterioration of financial conditions, in line with theoretical predictions.

High-frequency changes in interest-rate futures prices around Fed announcements offer a popular proxy for monetary policy shocks (Kuttner 2001; Gürkaynak et al. 2005b; Gertler and Karadi 2015). Financial market prices incorporate available information about the state of the economy and the systematic policy rule before the event. With a sufficiently narrow window that excludes other systematic news, the surprise, therefore, can be expected to include only the impact of Fed communication. However, the surprise will not be a pure measure of a monetary policy shock, i.e. a temporary deviation from a systematic policy rule, if market participants are imperfectly informed about the economy or the systematic policy. In either of these cases, the surprise will be contaminated by the systematic response of monetary policy to the state of the economy.

The CBI view focuses on Fed communication about the economic outlook. It ar-

gues that such communication moves private expectations either because the Fed has private information; uses different economic models to process public information available in the market (Melosi 2017; Nakamura and Steinsson 2018; Jarociński and Karadi 2020; Miranda-Agrippino and Ricco 2021); or because the Fed communication coordinates private expectations (Morris and Shin 2002). A CBI shock has two components, an underlying economic shock (e.g. a recent improvement in the health of the financial sector) and the central bank communication about it, which induces a sudden move in market perceptions. Jarociński and Karadi (2020) used high-frequency co-movement of interest rates and stock prices around FOMC statements to separately identify monetary policy (negative comovement) and CBI shocks (positive comovement). The idea behind the identification is that updates in private expectations about the economy due to a Fed communication needs to show up in high-frequency repricing of stock prices: depreciation in case of a policy tightening shock; and appreciation in case of a good news about the economy. The approach showed that a favourable CBI shock indeed causes a temporary boom in activity and prices in line with the view that the Fed announcement reveals an ongoing underlying boom that the policy partially counteracts and offsets within a couple of years. Correspondingly, the interest rate surprise cleaned from the impact of the CBI shock leads to a more pronounced downturn in activity and decline in prices than its uncleaned counterpart.

The FRN view (Bauer and Swanson 2023a,b) offers an alternative source of contamination of the monetary policy surprise. It also has two components. First, it argues that standard specifications that rely on monthly data, as those used in survey-expectation-revision equations by Campbell et al. (2012); Nakamura and Steinsson (2018), or in VAR models by Gertler and Karadi (2015) and Jarociński and Karadi (2020) omit some publicly available information on the concurrent economic shocks. Second, it argues that since the early 1990s the market has systematically underestimated the aggressiveness of systematic monetary policy, and shows that publicly available news indeed forecast a significant portion of the high-frequency interest rate surprises. Overall, the view argues that interest rate surprises caused by an unexpectedly aggressive response to fresh public news (FRN shocks) could be mistakenly identified as CBI shocks. To control for the impact of FRN shocks, the view suggests to purge high-frequency interest-rate surprises from the component that is predicted by fresh public news. A monetary policy shock thus identified generates an amplified economic downturn in a monthly VAR setting. Furthermore, the interest-rate surprise component predicted by fresh public news predicts a similar temporary boom as the CBI shock.

Which view is more relevant in the data? And how should one best disentangle monetary policy surprises and estimate their components? This paper sets out to address these questions in a unified framework.

The first contribution of the paper is to propose a methodology to simultaneously identify a monetary policy shock, a central bank information shock and a Fed response to news shock. To achieve this, it combines the predictability of interest rate surprises

with fresh public news with the high-frequency co-movement of interest rates and stock prices. The starting point in the identification is the requirement in the FRN view: an FRN shock needs to co-move with the component of the interest rate surprises that are explained by fresh public news. The key contribution of this paper is to point out a further identifying restriction inherent in the FRN view. In particular, a tighter than expected systematic policy necessarily leads to a stock price *depreciation* on impact. The argument is similar to an adverse monetary policy shock: the tighter than expected interest rate policy generates a temporary downturn - reducing the near-term cash flow of equities - and an increase in the discount rate. This restriction also helps to distinguish it from the CBI shock, which requires a *positive* comovement between interest rates and stock prices: good news about the economy leads to both a stock price appreciation and an accompanying tightening of systematic policy to offset its effects. Lastly, the monetary policy shock is identified as a shock that both co-moves with the interest rate surprises that are unexplained by fresh public news and generates a negative high-frequency comovement between interest rates and stock prices.

Its second contribution is to extend the sample of Fed events from FOMC statements used by previous research, to non-FOMC events, including Fed chair speeches, publication of minutes, etc., which have been shown to raise the relevance of the high-frequency interest-rate surprises (Swanson and Jayawickrema 2023; Swanson 2023, see also Altavilla et al. 2025 for the ECB).

The third contribution of the paper is to improve the precision of the estimates by utilizing the differences in the prevalence of the three channels in FOMC versus non-FOMC events. In particular, it combines sign restrictions with a heteroskedasticity-based identification (Rigobon 2003). It shows that the forecastability of the interest rate surprises is weaker at non-FOMC events than at FOMC events, indicating the FRN shocks are weaker during the former than during the latter. Furthermore, it shows that the negative relation between interest rate surprises is stronger during FOMC events than during non-FOMC events, suggesting that CBI shocks, which generate the positive comovement, are more prevalent in the non-FOMC subsample. This heteroskedasticity provides extra information that sharpens the inference.

We first assess the impact of the identified shocks on daily financial variables, such as stock prices, breakeven inflation, corporate bond spreads, and the VIX through local projections over a 60-day horizon. We find that central bank information shocks are a robust feature of the data even when FRN shocks are taken into account. A favorable CBI shock leads to increases in stock prices and breakeven rates and significant decreases in corporate bonds spreads and the VIX. We also find that FRN shocks exert a significant impact on financial prices. Notably, the responses to hawkish FRN shocks are *contractionary*, leading to significant declines in breakeven inflation rates and stock prices and increases in corporate bond spreads and the VIX. This is consistent with the FRN view: revisions in market perception about a more aggressive systematic policy indeed exert a qualitatively similar influence on financial variables as mone-

tary policy tightening shocks, albeit monetary policy shocks die out somewhat faster. Note that in daily regressions asset prices measured before the Fed announcement already incorporate the impact of recent public news. Therefore, changes relative to these prices caused by the FRN shock are free from the impact of these public news. Therefore the impact of FRN shocks in the daily regressions measure only the impact of the policy tightening. This is different from monthly regressions, where the FRN shock measures the combined impact of (i) a tighter than expected systematic policy and (ii) the impact of fresh public news, as we explain below.

Second, we implement the identification in a Bayesian VAR framework with monthly data between 1988:01-2025:03. The VAR includes the 1-year Treasury rate as a monetary policy indicator; a monthly interpolation (Stock and Watson 2010) of real GDP and the GDP deflator, as indicators for activity and prices; and the S&P 500 stock market index and the excess bond premium (Gilchrist and Zakrajsek 2012; Favara et al. 2016) as indicators for the financial conditions. The identification utilizes three high-frequency variables: the interest rate surprises (i) explained and (ii) unexplained by fresh public news and (iii) surprise in the S&P500 index, all in a 30 minute window around Fed announcements. They are measured in over 600 Fed communication events including FOMC statements, Fed-chair speeches, publication of minutes. We show robustness to restricting the event sample to FOMC statements to foster comparability with previous research.

We establish several results. First, we show that central bank information effects are a robust feature of the data. Even when we allow for the presence of Fed response to news effects, the identified CBI shocks cause a significant temporary boom in monthly data in all our specifications. Second, in our baseline monthly VAR specification with a wide range of Fed events, Fed response to news shocks have only marginal impact on monthly fluctuations. This is consistent with the view that the two components of the FRN shocks: (i) the stimulative impact of favorable fresh public news and (ii) the contractionary impact of tighter than expected systematic monetary policy broadly offset each other. One could obtain somewhat less sharp, but similarly valid results relying on simpler methods that ignore the presence of FRN effects (Jarociński and Karadi 2020). Considering only FOMC statements would be insufficiently informative to precisely estimate the effects of the three shocks. Third, we find that monetary policy shocks purged from the impact of both CBI and FRN effects generate a temporary downturn in activity and prices and are accompanied by a temporary deterioration of financial conditions, in line with theoretical predictions.

Related literature. Our paper contributes to the strand of empirical research that measures the causal impact of monetary policy on the economy through identifying temporary deviations from systematic policy - monetary policy shocks (Romer and Romer 2004; Christiano et al. 2005; Uhlig 2005; Arias et al. 2021). It builds on the insight of Gertler and Karadi (2015), who use high-frequency financial market surprises

around FOMC announcements (Kuttner 2001; Gürkaynak et al. 2005b) as instruments in a structural vector autoregression. Accurate high-frequency identification requires refinements if there is imperfect information about systematic policy or the state of the economy, which market participants learn about from policy announcements. Researchers have used different approaches to clean monetary policy shocks from these confounding factors. One strand cleans the surprises from the Fed's private information (Barakchian and Crowe 2013; Gertler and Karadi 2015); from a combination of the Fed's private information and public information (Miranda-Agrippino and Ricco 2021); or only from public information (Bauer and Swanson 2023a,b). Another strand of the literature achieves identification by looking at the co-movement with a wider set of high-frequency financial variables, including stock prices, (Cieślak and Schrimpf 2019; Jarociński and Karadi 2020) or text-based indicators (Acosta 2023). The literature has consistently found that confounding factors cause a significant bias, and controlling for them leads to monetary policy effects that are stronger and consistently in line with theoretical predictions. The paper's contribution is to provide a new measure of monetary policy shocks that cleans them from a wide set of confounding factors.

The paper also contributes to the debate on the existence of the central bank information effect: whether Fed communication about the economic outlook drives private expectations and the macroeconomy. Proponents argue that the Fed's efforts in macroeconomic data processing and analysis provide it relevant private information that private agents find useful (Romer and Romer 2000; Melosi 2017; Nakamura and Steinsson 2018). Those challenging the channel argue that the market efficiently aggregates information about the state of the economy, to which the Fed cannot usefully contribute; if the Fed has private information about anything, it is the nature of its systematic policy (Faust et al. 2004). Our paper revisits the debate and seriously considers whether the Fed response to news channel proposed recently by Bauer and Swanson (2023a,b) and outlined above is a valid alternative to the central bank information effect based on high-frequency identification (Jarociński and Karadi 2020; Miranda-Agrippino and Ricco 2021; Golez and Matthies 2025).

The paper extends the set of Fed communication events from FOMC announcements to include also press conferences, chair speeches, and publication of meeting minutes. Our paper shows that extending the set of events is not only important for improving the relevance of high-frequency surprises as shown by Swanson and Jayawickrema (2023) as a response to a critique of Ramey (2016); but also helps to cleanly assess the overall importance of different communication channels. Our paper shows that while on a narrow set of FOMC meeting announcements, Fed response to news channel is relevant in monthly VARs, its contribution becomes marginal in the broad sample of Fed communication events.

Finally, our paper contributes to the growing line of research that combines sign restrictions with statistical identification in vector autoregressions (Drautzburg and Wright 2023; Carriero et al. 2024; Andrade et al. 2025). Heteroskedasticity-based identi-

fication helps to improve the precision of the set-identified results obtained by sign restrictions. Conversely, sign restrictions help give structural interpretation to the shocks identified by heteroskedasticity, which, by itself, is only a statistical approach assigning no labels to the identified shocks.

Structure. The paper is structured as follows. Section 2 presents a unified theoretical framework with a precise decomposition of monetary policy surprises into a monetary policy, a Fed response to news and a central bank information shocks, and use the framework to derive the necessary restrictions for the identification of the three shocks. Section 3 presents the data, the construction of Fed event windows and reproduces the Bauer and Swanson (2023a,b) regressions. Section 4 presents the dynamic effects of the shocks on financial and macroeconomic variables within a local projection event study and a Bayesian VAR framework. Section 6 show the robustness of the results and Section 7 concludes. In Appendix A, we confirm that our baseline results are broadly consistent with the predictions of a standard macro-finance model.

2. Theoretical framework

This section develops a theoretical framework that distinguishes between three sources of monetary policy surprises: (i) pure policy shocks, (ii) market misperception of the Fed policy rule (the Fed response to news effect), and (iii) different views on the state of the economy between the Fed and markets (the Fed information effect). We show that neglecting (ii) and (iii) biases standard estimates and we propose a strategy to separately identify the three effects using high-frequency stock price responses.

2.1. Model setup and shock decomposition

Fed policy rule and market expectations. Federal Reserve sets its policy rate i as a function of the state of the economy, represented by x (for instance, the output gap). We assume the following policy rule:

$$i_t = \alpha^F x_t^F + \varepsilon_t \tag{1}$$

where α^F is the Fed's response coefficient, x^F is the Fed's assessment of the economy, and ε represents a pure monetary policy shock. The Fed announces its policy at the end of period t.

The market forms expectation about the to-be-announced policy rate just before the announcement, denoted $E^M_{t-\delta}(i_t)$:

$$E^M_{t-\delta}(i_t) = \alpha^M x_t^M \tag{2}$$

where α^M is the market's perception of the Fed's reaction function coefficient, and x^M is the market's view of the economy. These may differ from the Fed's true coefficient

and assessment.

The monetary policy surprise (MPS) is defined as the difference between the actual policy rate and the expected policy rate:

$$MPS_t = i_t - E^M_{t-\delta}(i_t)$$

Decomposing the monetary policy surprise. Subtracting (2) from (1) and rearranging yields a decomposition of the monetary policy surprise into three distinct components (suppressing time subscripts):

$$\underbrace{i - E^M(i)}_{MPS} = \underbrace{\alpha^F(x^F - x^M)}_{\text{CBI shock}} + \underbrace{(\alpha^F - \alpha^M)x^M}_{\text{FRN shock}} + \underbrace{\varepsilon}_{\text{MP shock}} \tag{3}$$

This decomposition reveals that markets can be surprised by the Fed's interest rate decision for three reasons. First, an unpredictable pure monetary policy shock (ε) can occur. Second, the market's assessment of the economy may differ from the Fed's assessment ($x^F \neq x^M$), creating what we call a *Central Bank Information* (CBI) shock. Third, markets may misperceive the Fed's policy rule ($\alpha^F \neq \alpha^M$), creating what we call a *Fed Response to News* (FRN) shock. We explain the naming convention below after we specify the information structure of the economy.

Information structure and economic dynamics. We specify the evolution of the state of the economy as follows:

$$x_t = \rho x_{t-1} - \theta i_{t-1} + \underbrace{\eta_t}_{\text{public news}} + \underbrace{\eta^F_t}_{\text{CB information}} \tag{4}$$

Economic fluctuations are driven by two types of i.i.d. shocks. The first component, η_t (public news), is observed by both the market and the Fed. The second component, η^F_t (Fed private information), is observed only by the Fed. This asymmetric information gives rise to the central bank information effect, reflecting the Fed's superior information about the economy or better ability to process dispersed economic information. Parameter $\rho \in [0, 1]$ determines the persistence of the state of the economy and $\theta > 0$ determines the impact of Fed policy rates on the economy.

This information structure directly connects to the shock decomposition in equation (3). When the Fed observes a favorable private information shock ($\eta^F_t > 0$), it raises rates based on information that the market lacks. The market is surprised, generating the CBI term: $\alpha^F(x^F_t - x^M_t) = \alpha^F \eta^F_t$. Hence the name "Central Bank Information shock"—it captures the Fed's response to information (η^F) that only the central bank possesses.

The FRN term arises from policy rule misperception. When favorable public news arrives ($\eta_t > 0$), both the Fed and the market observe it and update their assessment of the economy. However, if the market underestimates how aggressively the Fed

responds to this news ($\alpha^F > \alpha^M$), it will be surprised by the size of the rate hike. The FRN term $(\alpha^F - \alpha^M)x^M$ captures this surprise. We call it the "Fed Response to News shock" because it reflects the portion of the Fed's response to real-time economic news (η) that the market fails to anticipate due to policy rule misperception.

To summarize, in the context of the information structure and economic dynamics postulated in (4) the interest rate surprise becomes

$$MPS_t = \alpha^F \eta_t^F + (\alpha^F - \alpha^M)(\rho x_{t-1} - \theta i_{t-1} + \eta_t) + \varepsilon_t \tag{5}$$

Two immediate implications. This simple framework has two immediate implications for the monetary policy analysis.

First, the MPS is contaminated by the economic shocks η_t^F and η_t through the CBI and FRN terms, respectively. This means that monetary policy surprises conflate pure policy shocks with the Fed's endogenous response to economic conditions.

Second, policy rule misperception implies that the MPS is predictable using publicly available economic indicators. Formally, a regression of MPS on indicators of economic activity known before the Fed announcement can have a positive R-squared. The reason is that the FRN term is a function of x_t^M, so any indicators related to lagged conditions (x_{t-1}, i_{t-1}) and real-time signals of η_t will have explanatory power. This predictability, documented by Bauer and Swanson (2023a,b), provides key evidence for policy rule misperception.

2.2. Bias in standard estimates and the identification of CBI vs FRN effects

Does this contamination bias standard estimates of monetary policy effects? And can one empirically distinguish the CBI and FRN effects? We answer both questions affirmatively and to provide intuition, we trace how each shock affects the economy in the short and in the longer run.

Forecast revisions and shock propagation. Consider a forecast revision of the future state (x_{t+1}) between periods $t-1$ and t. Such a forecast revision captures the change in the projected outlook, and describes how each contemporaneous shock ($\varepsilon_t, \eta_t^F, \eta_t$) affects next period's economic conditions.

$$E_t^M(x_{t+1}) - E_{t-1}^M(x_{t+1}) = \underbrace{-\theta\varepsilon_t}_{\text{MP}} + \underbrace{(\rho - \theta\alpha^F)\eta_t^F}_{\text{CBI}} + \underbrace{\underbrace{(\rho - \theta\alpha^F)\eta_t}_{\text{public news}} \; \underbrace{-\theta(\alpha^F - \alpha^M)E_{t-1}^M x_t}_{\text{policy rule misperception}}}_{\text{FRN}} \tag{6}$$

The decomposition reveals three distinct effects. First, a pure monetary policy shock ($\varepsilon_t > 0$) reduces the forecast revision proportionally with coefficient θ, the direct

impact of monetary policy on the economy. This is the key parameter researchers aim to estimate (see equation 4).

Second, a favorable central bank information shock ($\eta_t^F > 0$) raises the forecast revision under reasonable parameterizations $\rho - \theta\alpha^F > 0$.[1] The information shock directly improves economic conditions, and this positive effect dominates the contractionary impact of the Fed's policy response.

Third, the FRN shock has two components. Favorable public news ($\eta_t > 0$) raises the forecast revision, again under the same reasonable parameter values. However, if markets underestimate the Fed's response coefficient ($\alpha^F > \alpha^M$), the unexpectedly large rate hike partially offsets this improvement. The net effect on the forecast revision typically remains positive.

Comparing the impact of contemporaneous shocks on (6) and (5) we note that $\varepsilon_t > 0$ increases the MPS and decreases the forecast revision, but η_t and η_t^F generate the opposite co-movement: $\eta_t > 0$ and $\eta_t^F > 0$ increase both the MPS and the forecast revision (the increase of the MPS after $\eta_t > 0$ relies on $\alpha^F > \alpha^M$). This observation serves as the intuition for the formal result stated next.

The bias in local projections. Suppose a researcher estimates the effect of monetary policy on the future state of the economy using a local projection, with MPS_t as a proxy for the monetary policy shock. (Equivalent estimates can be obtained with a VAR, see Plagborg-Møller and Wolf 2021) The researcher estimates

$$x_{t+1} = \mu + \theta^{LP} MPS_t + \delta_x x_{t-1} + \delta_i i_{t-1} + \xi_t,$$

where x_{t-1}, i_{t-1} are standard controls, or equivalently

$$x_{t+1} = \theta^{LP} \overline{MPS}_t + \xi_t,$$

where $\overline{MPS}_t = \varepsilon_t + \alpha^F \eta_t^F + (\alpha^F - \alpha^M)\eta_t$ is the MPS orthogonalized with respect to the controls (x_{t-1}, i_{t-1}).[2]

It is straightforward to show that the population regression coefficient is

$$\begin{aligned} E(\theta^{LP}) = \frac{\operatorname{cov}(x_{t+1}, \overline{MPS}_t)}{\operatorname{var}(\overline{MPS}_t)} &= -\theta + \frac{\rho\alpha^F \operatorname{var}(\eta^F)}{\operatorname{var}(\overline{MPS})} + \frac{\rho(\alpha^F - \alpha^M)\operatorname{var}(\eta)}{\operatorname{var}(\overline{MPS})} \\ &= -\theta + \frac{\rho}{\alpha^F}\frac{\operatorname{var}(CBI)}{\operatorname{var}(\overline{MPS})} + \frac{\rho}{\alpha^F - \alpha^M}\frac{\operatorname{var}(\overline{FRN})}{\operatorname{var}(\overline{MPS})}. \end{aligned} \tag{7}$$

The researcher hopes to estimate the true effect $-\theta$ but instead obtains a biased estimate. The bias is unambiguously upward under plausible parameters (assuming $\alpha^F > \alpha^M$ as argued by Bauer and Swanson 2023a,b).

[1] This parameter restriction realistically requires that systematic monetary policy does not fully offset the impact of the central bank information shock on the outlook.

[2] The equality of θ^{LP} in the two regressions follows from the Frisch-Waugh-Lovell theorem.

The economic interpretation is clear: when the Fed observes favorable information shocks ($\eta_t^F > 0$) or responds aggressively to positive public news ($\eta_t > 0$), it raises rates and the economy subsequently improves. The researcher attributes these output improvements to the policy shock, thus underestimating the contractionary effect of policy. In other words, the regression coefficient θ^{LP} conflates the true effects of monetary policy with the effects of the shocks to which the Fed responds. Standard high-frequency approaches therefore underestimate the strength of monetary policy's impact on the economy.

Identification of FRN vs CBI: High-frequency stock price responses. How can we separately identify these confounding shocks? Our key insight is that CBI and FRN shocks have *opposite* effects on stock prices in narrow windows around Fed announcements, despite both raising interest rates. This sharp prediction enables identification.

Stock price responses to policy surprises. To formalize this intuition in our stylized model, we define a 'stock' price s_t as the discounted value of next-period outlook:

$$s_t = E_t^M \left[x_{t+1} - i_t \right],$$

where the outlook correlates with future cash flow. The high-frequency change in the stock price—the change between just before and just after the Fed announcement—can be decomposed as follows:

$$s_t - s_{t-\delta} = E_t^M(x_{t+1}) - E_{t-\delta}^M(x_{t+1}) - MPS_t = \rho\eta_t^F - (1+\theta)MPS_t = \underbrace{(\rho - (\theta+1)\alpha^F)\eta_t^F}_{\text{CBI}} \underbrace{-(1+\theta)(\alpha^F - \alpha^M)E_{t-\delta}^M(x_t)}_{\text{FRN}} \underbrace{-(\theta+1)\varepsilon_t}_{\text{MP}} \quad (8)$$

Contrasting high-frequency responses. The three shocks have distinct high-frequency effects on stock prices:

- *CBI shock ($\eta_t^F > 0$):* Stock prices *rise* under reasonable parameters ($\rho > (\theta+1)\alpha^F$). There are two opposing forces: the revelation of positive private information leads markets to revise up their outlook by $\rho\eta^F$, but the accompanying rate hike ($\alpha^F\eta^F$) reduces both future activity and its present discounted value. The positive information effect dominates when the CBI shock's impact on the outlook is not fully offset by monetary policy's combined effect on growth and the discount rate. Empirically, Jarociński and Karadi (2020) show that interest rate surprises associated with stock price appreciation are indeed followed by stronger economic activity, consistent with this channel.
- *FRN shock ($\eta_t > 0$ with $\alpha^F > \alpha^M$):* Stock prices *fall* unambiguously. Although positive public news η_t improves fundamentals, this information is already incorporated into prices at $t-\delta$ since markets observe it. The only *news* revealed in the announcement is that the Fed raised rates more than markets expected. This surprise reduces

expected future activity by θ and increases the discount rate, both depressing stock prices. There is no offsetting positive dividend effect in the high-frequency window, as that effect has already been priced in.

- *MP shock ($\varepsilon_t > 0$):* Stock prices *fall.* A pure contractionary monetary policy shock reduces expected activity and increases discount rates. This is the standard channel.

Crucially, this identification strategy exploits the narrow announcement window. At monthly frequencies, public news η_t affects both interest rates and stock prices directly, obscuring the distinction between FRN and CBI shocks. But in high-frequency data, only information revealed *in the announcement* matters. Since η_t is already known while η_t^F is not, the FRN shock moves rates and stocks in opposite directions, while the CBI shock moves them in the same direction.

We use these opposite high-frequency comovements to separately identify CBI and FRN shocks in our empirical analysis. Note also that the absence of a direct positive public news effect similarly applies to other high-frequency asset price responses to FRN shocks, an observation we exploit in Section 4.3 to interpret impulse responses in daily regressions.

Extension: time-varying policy rule. One might object that markets should learn the policy rule over time, eliminating systematic misperception. A more realistic framework allows the Fed's response coefficient to evolve as a random walk:

$$\alpha_t^F = \alpha_{t-1}^F + u_t. \tag{9}$$

In the appendix we show that in this framework the FRN shock remains present in the MPS, but with two important differences from the baseline model: (1) it becomes largely indistinguishable from the pure monetary policy shock ε and does not lead to bias in estimates of monetary policy effects; (2) it does not imply predictability of the MPS using public information. Predictability relies only on the average misperception persisting throughout the sample (*average* $\alpha^F - \alpha^M > 0$), while fluctuations of this quantity are irrelevant for predictability.

3. Data

This section explains our data on high-frequency market reactions to Fed communication (high-frequency surprises). First, we replicate and extend the Gürkaynak et al. (2005a) dataset of high-frequency surprises, adding also surprises around more types of Fed communication, such as Fed chair speeches and minutes releases, following Swanson and Jayawickrema (2023). Second, we inspect the co-movement of interest rate and stock price surprises in this dataset and run the Bauer and Swanson (2023b) predictive regressions with the interest rate surprises.

3.1. Data sources

We take high-frequency financial market data from three sources: TickData, LSEG Tick History and Pi Trading.[3] The main variables of interest are Eurodollar/SOFR futures[4] and the S&P500 futures. The first four futures tenors, ED1, ED2, ED3, ED4, come from TickData (until 2004), and LSEG Tick History (2005-2022). From January 2022 onwards we use the matching SOFR futures (Acosta et al. 2024). We take the S&P 500 futures from TickData (until 2008), from 2008 onwards we use E-mini S&P 500 futures from LSEG Tick History.[5] Relying on S&P 500 futures, as opposed to the actual stock index, has the advantage that the futures market is active while the New York Stock Exchange is closed.

We construct the interest rate surprise, MPS, as the first principal component of the ED1, ED2, ED3 and ED4 surprises computed over Fed event windows defined below.

3.2. Events, samples

We construct event windows in two steps. First, we collect the timestamps of Fed events and define the time windows that bracket those events. Second, we collect the timestamps of the potential interfering events and adjust the Fed event windows to remove the overlap with the interfering events.

Fed event windows. We collect the timestamps of FOMC announcements, post-FOMC press conferences, Fed chair speeches and congressional testimonies,[6] and the release of the FOMC meeting minutes. From 1997 onward, the source is the Bloomberg Economic Calendar and the Federal Reserve website. Before 1997 we took the times of FOMC announcements from Bauer and Swanson (2023a) and the times of Fed chair speeches and testimonies from Board of Governors of the Federal Reserve System and Greenspan (2019). Table 1 reports the count of events of each type. The 357 FOMC announcements include FOMC press releases (from 1994 onward) as well as timestamps of the first post-FOMC open market operations collected by Bauer and Swanson (2023a) (before 1994). The second column reports the assumed duration of each type

[3] We have access to LSEG Tick History database with real-time updates, but the dataset starts only in 1996. Therefore, we combine it with data from TickData (access to time series till 2019) and Pi Trading, which both go back to before 1988, when we start our analysis. We combine the datasets in a way to maximize coverage from a single source. Surprises calculated from different sources using overlapping samples are almost identical.

[4] Eurodollar futures were based on the London Interbank Offered Rate (LIBOR). LIBOR was assembled using rates on unsecured interbank transactions. The rate and derivatives based on it were phased out between 2021-2023. It was gradually replaced by the Secured Overnight Financing Rate (SOFR), a rate based on transactions in the U.S. Treasury repurchase (repo) market.

[5] E-mini S&P500 futures became the most liquid market due to its smaller contract size. The dataset contains also Fed funds futures and Treasury futures which we only use in the robustness exercises and report in more detail in the Appendix.

[6] In some cases (10 out of 369), we know that the text of the congressional testimony was released ahead of the actual testimony. We treat the release of the text of the congressional testimony as a separate event.

TABLE 1. Fed events, January 1988 - March 2025: counts, window lengths, median absolute ED3 surprises, and the number of large surprises

Event type	Count	Duration (min)	Med(|ED3|)	#|ED3| > 3bp
FOMC Announcement	357	0	0.020	158
Non-FOMC Events	1256	-	0.010	339
of which				
Chair Speech	576	90	0.010	135
Minutes	217	30	0.010	35
Press Conference	81	60	0.015	29
Testimony	372	180	0.015	138
Testimony Release	10	30	0.000	2

Sample January 1988 - March 2025

of event. We define windows around the events that start 10 minutes before the start of the event and end 20 minutes after the end of the event.

Interfering events. Asset prices in Fed windows might be affected by some other interfering events that occur simultaneously. Therefore, we adjust the Fed event windows for the interfering event windows. Kerssenfischer and Schmeling (2024) systematically study which types of events move financial markets the most, and we choose interfering events based on their results. In particular, we use the set of 18 types of interfering events consisting of major US data releases, Treasury auction results,[7] and ECB monetary policy decisions and press conferences We create windows around these interfering events that start 10 minutes before and end 20 minutes after the interfering event. We then eliminate from Fed event windows any overlap with the interfering event windows. If an interfering event occurs in the middle of a longer Fed window and this procedure would leave us with two windows, we leave only the longer of the two windows. 263 windows (16%) are affected by the adjustment.

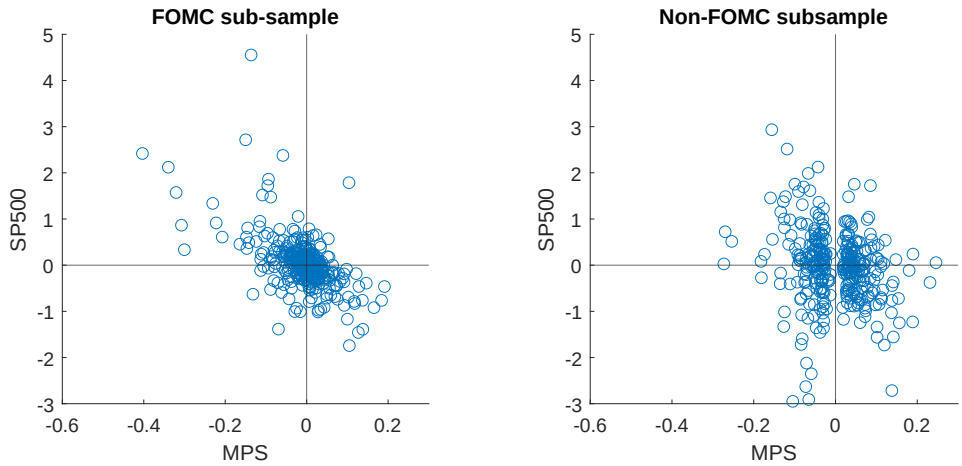
In the last step, we compute the changes of the main interest rates and of the S&P500 stock index in the Fed windows. The fourth column of Table 1 reports the median absolute change of ED3 (the third Eurodollar future).

We have verified that we are able to closely replicate the dataset of Gürkaynak et al. (2005b) using our procedures and data sources. We compare the 2019 version of that well-known dataset with the surprises that we obtain in the FOMC announcement windows. The correlations on an event-by-event basis are at least 94% for all variables.

We define big events as those during which the ED3 moves by at least 3 basis points (the same threshold is used by Bauer and Swanson 2023b). Given these data we create the following subsamples:

[7]These are all the US events that enter their Figure 1, including US Employment report, US ISM PMI Manufacturing, US Auction Result Notes, US Jobless Claims, US CPI and Earnings, US Chicago PMI, US Auction Result Bond, US University of Michigan Surveys, US Conference Board Leading Indicator, US Retail Sales, US Durable Goods, US ISM PMI Non-Manufacturing, US GDP, US Auction Announcement Bill, US PPI, US Existing Home Sales, US New Home Sales.

FIGURE 1. Scatter plots of interest rate (MPS) and stock price surprises (SP500) in FOMC and non-FOMC subsamples



The negative relation is stronger in the FOMC subsample suggesting that CBI shocks are more prevalent in the non-FOMC subsample.

- Baseline: FOMC announcements and "big" non-FOMC events, 696 observations
- FOMC announcements, 357 observations
- "Big" non-FOMC events, 339 observations

When estimating the impact of the shocks on other variables we drop all Fed events that happen in March 2020, in order to prevent the unusual Covid-related fluctuations from having an outsized impact on the results. We explain our decision in Appendix D.

3.3. Co-movement of interest rate and stock price surprises, Bauer-Swanson regressions

In this section, we inspect the features of the data related to the theoretical framework: the co-movement of interest rate and stock price surprises, and the predictability of interest rate surprises. We look separately at the FOMC and the non-FOMC subsamples.

Figure 1 plots the relationship between interest rate and stock price surprises, separately for the FOMC and non-FOMC subsamples.[8] The two scatterplots have interest surprises (MPS) on the horizontal axis and stock price surprises on the vertical axis. Each dot on the left scatterplot corresponds to an FOMC event and each dot in the right scatterplot corresponds to a non-FOMC event. We can see that the negative relation between interest rate surprises and stock price surprises is stronger in the FOMC subsample. In the non-FOMC subsample there are more cases of a positive co-movement. This suggests that the CBI shocks, which are the only ones that generate the positive co-movement, are more prevalent in the non-FOMC subsample.

Next, we replicate Bauer-Swanson regressions of the monetary policy surprises (MPS) on observable economic variables and decompose MPS into a fitted value (FIT)

[8] In this figure as in most of the paper we focus by default on the "big" non-FOMC events. Figure A5 shows analogous plots with all non-FOMC events.

TABLE 2. Explanatory variables of the Bauer-Swanson regressions

Acronym	Definition	Used in[a]
s_unemplrate	Unemployment rate surprise	AER, IMF
s_nfp	Non-farm payrolls surprise	AER, MA, IMF
s_rgdp	Real GDP surprise	AER, IMF
s_corecpi	Core CPI surprise	AER, IMF
m_corecpi	Core CPI median forecast	AER
dcorecpi6m	6-month change in Core CPI	AER
cycle_bbk	BBK Index	AER
l1l90logsp500	Δ log SP500 (3m)	AER, MA, IMF
l1l90pc2	Δ yield curve slope (3m)	AER, MA
l1l90logbcomsp	Δ log commodity price (3m)	AER, MA, IMF
isk1m	Treasury yield skewness (1m)	MA
employ_yoy	Employment growth (12m)	MA
l1l90sveny02	Δ 2-year Treasury (3m)	IMF
l1l90sveny10	Δ 10-year Treasury (3m)	IMF
l1l90baa10y	Δ Baa spread (3m)	IMF
d3mwuxia	Δ Shadow fed funds rate (3m)	IMF
l1l30anfci	Δ Chicago Fed NFCI (1m)	IMF

[a] AER denotes Bauer and Swanson (2023b), MA denotes Bauer and Swanson (2023a) and IMF denotes Swanson (2024).

and the residual (RES).

$$MPS_t = \underbrace{\beta y_t}_{\text{FIT}} + \underbrace{\varepsilon_t}_{\text{RES}}, \tag{10}$$

where MPS is an interest rate surprise and y_t is a vector of variables known to the market participants before the Fed event window. Bauer and Swanson use a different vector y_t in the alternative papers (Bauer and Swanson 2023b,a; Swanson 2024). In order to be conservative, we take as y_t the union of all readily available regressors from their three recent papers.

Table 6 shows that on FOMC meeting days, which was used as a baseline sample in Bauer and Swanson (2023a,b), the regressors indeed predict a sizable fraction, around 19% of the surprises - similarly to the results emphasized by Bauer and Swanson. However, when we extend the events to other type of Fed events, the explanatory power of the same set of regressors declines substantially. The R-squared drops to single digits in the non-FOMC sample and in the combined sample.[9]

[9] We are confident that this result is robust and not due to an omission of some important regressors, as we are using a very comprehensive set of regressors. Also, Altavilla et al. (2025) report a similar finding for the ECB Board Member speeches.

TABLE 3. Bauer-Swanson regressions over the baseline, the FOMC and the non-FOMC samples

	Baseline	FOMC	Non-FOMC
(Intercept)	-0.011	-0.020*	-0.003
	(0.008)	(0.010)	(0.014)
Unemployment surprise	0.017	0.008	0.021
	(0.017)	(0.020)	(0.029)
Payrolls surprise	0.017	0.015	0.024
	(0.012)	(0.014)	(0.041)
Real GDP surprise	0.012**	0.005	0.017*
	(0.006)	(0.007)	(0.009)
Core CPI surprise	0.023	0.028	0.006
	(0.026)	(0.033)	(0.042)
Core CPI median forecast	-0.021	-0.016	-0.015
	(0.036)	(0.043)	(0.060)
6-month change in core CPI	-0.006	-0.002	-0.013*
	(0.004)	(0.005)	(0.007)
BBK index	-0.001	-0.002	-0.000
	(0.001)	(0.002)	(0.003)
Δ log S&P500 (3m)	0.070	0.136**	0.005
	(0.049)	(0.065)	(0.072)
Δyield curve slope (3m)	-0.073	-0.252	0.235
	(0.164)	(0.213)	(0.245)
Δ log pcommodity (3m)	0.110***	0.160***	0.050
	(0.039)	(0.053)	(0.058)
Treasury yield skewness (1m)	0.028**	0.029**	0.037*
	(0.012)	(0.014)	(0.020)
Employment growth (12m)	0.422*	0.519**	0.124
	(0.219)	(0.244)	(0.418)
Δ 2-year Treasury (3m)	-0.012	-0.074	0.092
	(0.050)	(0.065)	(0.074)
Δ 10-year Treasury (3m)	0.020	0.092	-0.101
	(0.063)	(0.082)	(0.094)
Δ Baa spread (3m)	-0.007	-0.014	0.018
	(0.011)	(0.014)	(0.018)
Δ shadow fed funds rate (3m)	-0.007	0.014	-0.015
	(0.008)	(0.011)	(0.012)
Δ Chicago Fed NFCI (1m)	0.060***	0.049*	0.039
	(0.017)	(0.025)	(0.026)
N	696	357	339
R-squared	0.060	0.180	0.064
Adj.R-squared	0.037	0.139	0.014

4. Dynamic effects

This section constructs the three shocks and tracks their effects in daily (yield curve, stock prices, TIPS/breakeven rates, spreads) and monthly data (activity, prices, financial variables).

4.1. Identification: sign and magnitude restrictions

After decomposing MPS as: MPS = FIT + RES, we specify the following sign and magnitude restrictions relating these variables to the underlying economic shocks.

a. Monetary policy shock (MP) is associated with a negative co-movement between the unexplained component of the interest rate surprise (RES) and the stock price surprise (SP500)
b. Fed response to news shock (FRN) is associated with a negative co-movement between the predicted component of the interest rate surprise (FIT) and the stock price surprise (SP500)
c. Central bank information shock (CBI) is associated with a positive co-movement between the interest rate surprise (MPS) and the stock price surprise (SP500)
d. "Good proxy" restriction: RES is a good proxy for the monetary policy (MP) shock, in the sense that the MP shock impacts the RES more than the FIT.
e. "Good proxy" restriction: FIT is a good proxy for the Fed response to news (FRN) shock, in the sense that the FRN shock impacts the FIT more than the RES.

These restrictions can be represented as follows:

$$\begin{pmatrix} RES \\ FIT \\ SP500 \end{pmatrix} = \begin{pmatrix} + & c_{12} & c_{13} \\ c_{21} & + & c_{23} \\ - & - & + \end{pmatrix} \begin{pmatrix} MP \\ FRN \\ CBI \end{pmatrix} \tag{11}$$

where - denotes a negative sign restriction, + denotes a positive sign restriction, and

$$c_{13} + c_{23} > 0, \qquad \text{Assumption c} \tag{12}$$

$$|c_{21}| < c_{11}, \qquad \text{Assumption d} \tag{13}$$

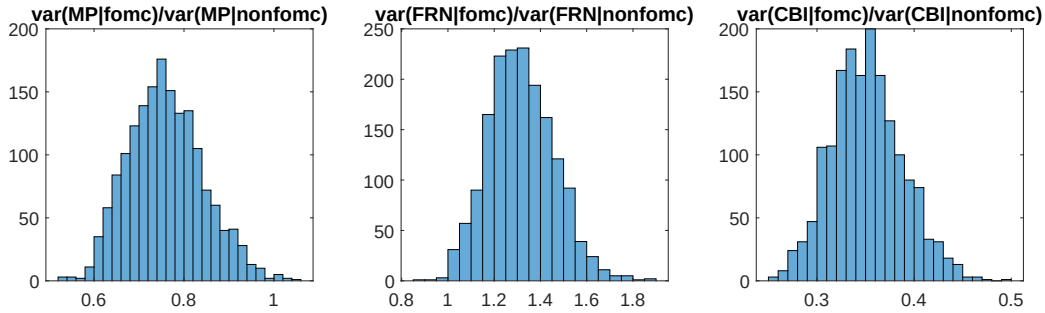
$$|c_{12}| < c_{22}, \qquad \text{Assumption e.} \tag{14}$$

In a nutshell, the sign restrictions reflect the theoretical analysis in Section 2 while the magnitude restrictions state that if RES is a good proxy for MP and FIT a good proxy for FRN, then c_{12} and c_{21} will be small.

4.2. Sharpening the inference through heteroskedasticity-based identification

Sign and magnitude restrictions on their own are consistent with many different decompositions and consequently do not deliver very sharp conclusions on the effects

FIGURE 2. Ratios of shock variances across the two subsamples



The figure shows that the mix of shocks changes across the two subsamples. Monetary policy shocks are somewhat smaller in the FOMC subsample , FRN shocks are somewhat larger in the FOMC subsample, whereas CBI shocks are much larger the non-FOMC subsample.

of different shocks. We report these results in Section 5. However, we can sharpen the inference by bringing in the additional information contained in the heteroskedasticity between the FOMC and non-FOMC events.

Two observations suggest that the mix of shocks in the FOMC and non-FOMC events is significantly different. The first observation comes from Figure 1: in the non-FOMC subsample there are more cases of a positive co-movement between interest rate and stock price surprises. This suggests that the CBI shocks, which are the only ones that generate the positive co-movement, are more prevalent in the non-FOMC subsample. The second observation on heteroskedasticity comes from Table 6: the R-squared of the Bauer-Swanson regressions is much higher in the FOMC subsample than in the non-FOMC subsample. This suggests that FRN shocks, proxied by the fitted values of the Bauer-Swanson regressions, are more prevalent in the FOMC subsample.

To exploit the information contained in the heteroskedasticity of our data, we estimate the following model generalizing equation (11):

$$y_t = C' u_t, \qquad u_{nt} \sim \mathcal{N}(0, \sigma^2_{ns}), \qquad s \in \{\text{FOMC, Non-FOMC}\} \tag{15}$$

where t indexes the Fed events, $y_t = (RES_t, FIT_t, SP500_t)$ are the residual and fitted value of the Bauer-Swanson regression and the stock price surprise observed at event t, $u_t = (MP_t, FRN_t, CBI_t)$ are orthogonal MP, FRN and CBI shocks occurring at event t and C collects the effects of the shocks u_t on the observable quantities y_t. The key feature of the model is that shock n ($n \in \{\text{MP,FRN,CBI}\}$) has a different variance on the FOMC and on non-FOMC events. The fact that C is identified in the presence of such heteroskedasticity is known since Rigobon (2003), (see also, e.g., Lanne et al. 2010; Brunnermeier et al. 2021). We estimate model (15) with Bayesian methods, using a Gibbs sampler described in detail in the Appendix. When sampling C and u_t, $t = 1, \ldots T$ we impose the sign and magnitude restrictions stated in Section 4.1. About 17 percent of the draws satisfy these sign restrictions.

Figure 2 reports the posterior distributions of the ratios of shock variances across regimes. Heteroskedasticity-based identification only works if on at least two of the

shock variances change across the two subsamples (variance ratios must be different from one) and we can see that the variances do change (variance ratio of one is a very low probability event). The CBI shocks are much larger in the non-FOMC sample, the MP shocks are somewhat larger in the non-FOMC sample, and the FRN shocks are larger in the FOMC subsample.

TABLE 4. Variance decompositions in the estimated heteroskedastic model (15).

	RES	FIT	STOCK
MP	**0.81**	0.06	0.43
	(0.10)	(0.05)	(0.14)
FRN	0.04	**0.93**	0.09
	(0.03)	(0.05)	(0.05)
CBI	0.15	0.01	0.48
	(0.09)	(0.01)	(0.14)

Posterior mean, standard deviation in parentheses.

The estimated C implies that 81% of the variance of the Bauer-Swanson residuals reflect monetary policy shocks, 4% reflect FRN shocks and 15% CBI shocks. Bauer-Swanson fitted values are mainly reflecting the FRN shocks (93% of the variance), with some contamination by monetary policy shocks (6%) and very little role for the CBI shocks.

4.3. Event study: Impact of shocks in daily data

Figure 3 reports local projections of daily financial variables on the raw interest rate surprise (MPS), and the posterior median MP, FRN and CBI shocks constructed in the previous section. More in detail, we run the regressions of $y_{t+h} - y_{t-1} = \alpha + \beta_h u_t + \varepsilon_t^h$, where y is a response variable, u is a shock, t is the day of the Fed event, h is the horizon, in days. β_h captures the effect of the shock at horizon h. We have also run specifications with additional lagged controls and obtained very similar results. All shocks are reported with the sign that leads to a positive interest rate surprise. These correspond to a contractionary MP shock, and favorable FRN and CBI shocks, the latter two associated with good news about the economy.

Two results stand out. First, the impact of the favorable FRN shock is contractionary. Similarly to the MP shock, it leads to a depreciation of the stock prices (logsp500), a decline in the inflation expectations (1-year inflation swap rate, uscpiz1y; 10-year breakeven inflation rate, bkeven10), and a worsening of financial conditions, as reflected in the increase in the corporate bond spreads (ice us hyld oas) and VIX (vixcls). These results align well with the narrative that, at a daily frequency, the favorable public news is mostly incorporated in asset prices by the Fed announcement date, and the surprise associated with the more aggressive systematic response to this news amounts to a contractionary policy shock. Notably, the interest rate response to the FRN shock (sveny02,sveny10) is somewhat more gradual and persistent than the response to the

MP shock and leads to a more pronounced impact on the inflation expectations. This is in line with changes in systematic policy, which leads to more persistent changes in policy and a larger cumulative impact.

Second, the effects of the CBI shock are different from those of the other two shocks. Financial conditions improve and long-term breakeven rates increase. Most notably, the long-term interest rates show a significantly different dynamics – with implications to the post-FOMC drift documented by Brooks et al. (2018). The figure suggests that the post-FOMC drift in Treasury yields is primarily driven by CBI shocks, with FRN shocks also contributing to a lesser extent. In contrast, monetary policy shocks that have been adjusted to exclude the effects of CBI and FRN do not produce any such drift in the 10-year Treasury rate.

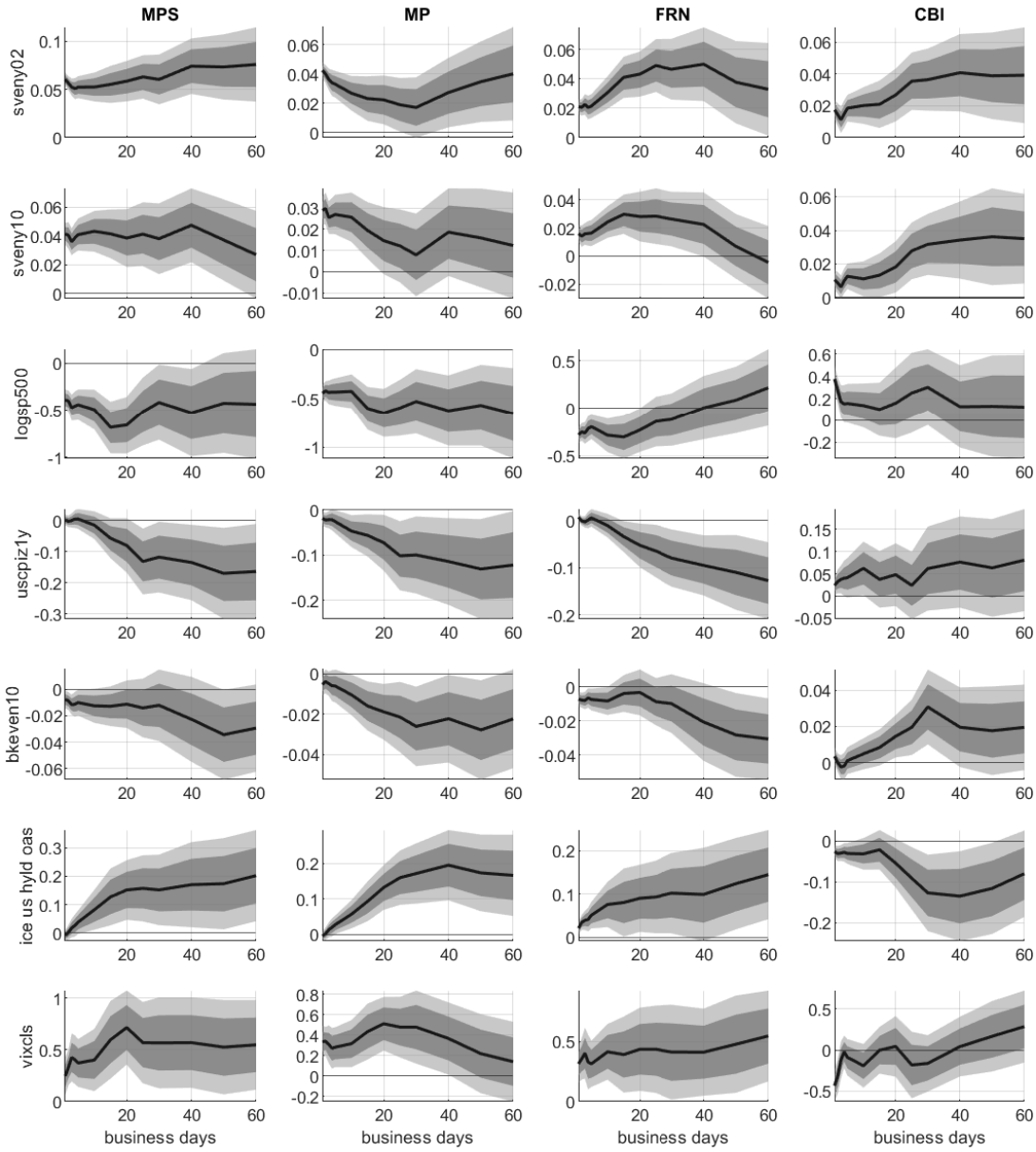
4.4. Macroeconomic effects: monthly VAR

We now turn to assessing the dynamic effects of the three shocks in a monthly Bayesian VAR framework. Our baseline VAR includes eight variables: three shocks identified in the high-frequency and five low-frequency macroeconomic variables. Low-frequency variables include the monthly 1-year Treasury rate as a monetary policy indicator (gs1); the logarithm of the monthly average S&P500 stock price index (logsp500); monthly interpolations of the GDP (us gdp) and the GDP deflator (us gdpdef) as indicators of real activity and price level; and, finally, the Gilchrist and Zakrajsek (2012) external bond premium (ebp) as an indicator of financial conditions.

The VAR has 12 lags. The sample is monthly, from January 1988 to March 2025. The estimation is Bayesian with a standard specification of the Minnesota prior (with the "overall tightness coefficient" of 0.2). The posterior simulation accounts for the uncertainty about the shock identification, i.e. we repeatedly draw the shocks from their posterior distribution implied by model (15) and then we draw the VAR parameters conditionally on the draw of the shocks. We report the results based on 2,000 draws from the posterior.

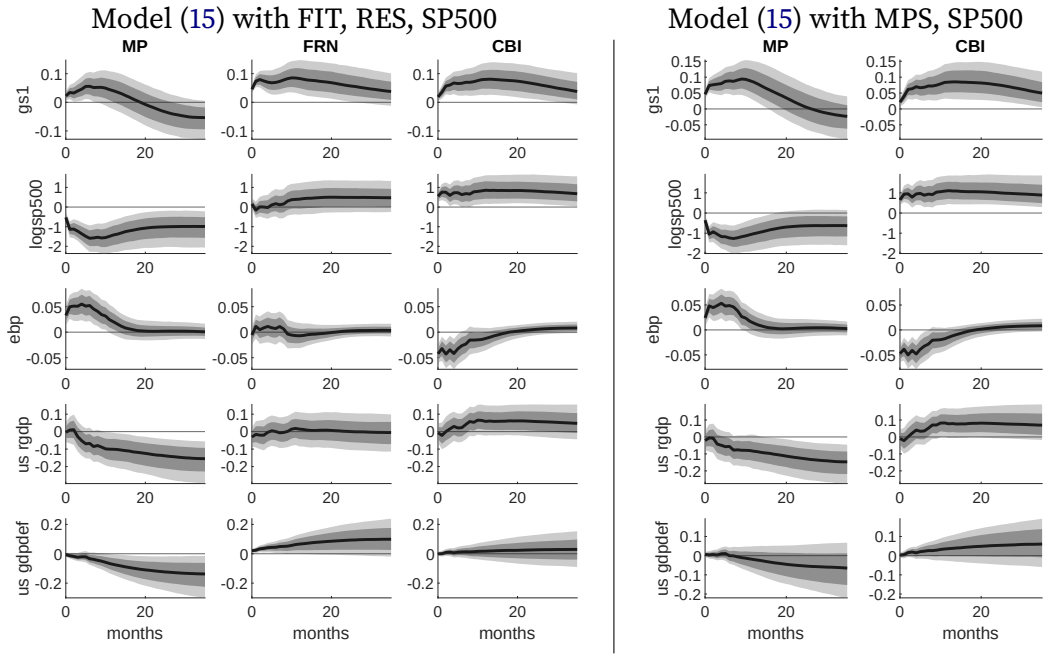
Figure 4 presents the impulse responses to the monetary policy (MP), Fed response to news (FRN) and central bank information (CBI) shocks identified as outlined in Sections 4.1 and 4.2. The results make two observations clear. First, the Fed response to news shock while significantly tightens the monetary policy indicator for a few months, leads to no significant response in either activity or financial conditions in the following months (while the response of prices is significant). This lack of a broad impact is consistent with the stimulative effects of favorable fresh public news and the contractionary effects of tighter than expected systematic monetary policy broadly offsetting each other. Second, central bank information shocks exert a significant dynamic impact on financial conditions and improve activity (Jarociński and Karadi 2020), albeit only marginally significantly. This happens even after we control for the potential presence of the Fed response to news channel. The central bank information shock is significantly different from the impact of the monetary

FIGURE 3. Daily local projections, $y_{t+h} - y_{t-1} = \alpha + \beta_h u_t + \varepsilon_t^h$



Point estimates of the response (β_h), with heteroskedasticity robust 68% and 90% bands. Responses to raw interest rate surprises (MPS), monetary policy (MP) shocks, Fed response to news (FRN) shocks, and central bank information (CBI) shocks. The figure highlights that while all three shocks raise persistently the 2-year (sveny02) and 10-year (sveny10) interest rates, MP and FRN shocks lead to a contractionary impact on stock prices (logsp500) and inflation expectations (1-year inflation swap rate, uscpiz1y; and 10-year breakeven rate, bkeven10), and a deterioration of financial conditions (corporate bond spreads, ice us hyld oas; market volatility, vixcls), CBI shocks reflect positive news about economic fundamentals and lead to favorable financial conditions.

FIGURE 4. Dynamic effects of MP, FRN and CBI shocks in a monthly VAR. Heteroskedasticity-based identification.



The results indicate that FRN shocks tighten 1-year Treasury rates (gs1) but have limited effects on activity (us rgdp), prices (us gdp def), and financial conditions (excess bond premium, ebp), while CBI shocks have a significant positive impact on the macroeconomy. A monetary policy tightening due to MP shocks leads to lower stock prices and worsening financial conditions, whereas a tightening driven by a favorable CBI shock results in improved stock valuations and financial conditions, with temporary increases in activity and prices. The shaded areas show 68% and 90% posterior bands. The black solid line shows the median impulse response. The VARs are monthly 1988:01-2025:03. Baseline set of Fed events (both FOMC and non-FOMC).

policy shock, so ignoring its presence leads to biased estimates. A monetary policy tightening caused by a monetary policy shock, in line with theoretical predictions, depreciates stock prices and leads to a worsening financial conditions, and causes a persistent downturn. In contrast, a tightening caused by a favorable central bank information shock appreciates stock valuations and improves financial conditions and causes a temporary upturn in activity and prices that persistently tight monetary policy offsets in one or two years.

4.5. Two simplified approaches

The above analysis simultaneously accounts for both central bank information (CBI) and Fed response to news (FRN) shocks. What are the consequences of ignoring either one of these confounding shocks?

CBI shocks only. We first account for the CBI shock only. We skip the step of running the Bauer-Swanson regressions and use directly the MPS and the stock price surprise. Figure 1 displays these data. With these data we estimate the heteroskedastic model (15). The estimation yields one shock accounting for the negative co-movement between

MPS and SP500, and the other one accounting for their positive co-movement. We label these shocks as MP and CBI respectively. The right panel of Figure 4 reports the dynamic effects of these two shocks in the same monthly VAR. The effects of both shocks are overall similar to the baseline case with three shocks. The main differences are that the monetary policy shock implies a larger and more persistent response of the 1-year Treasury yield and a smaller and less significant decline of the GDP deflator; and the CBI shock now significantly improves activity, and raises the price level, albeit only marginally significantly.

Fed response to news only. Second, we account for the Fed response to news only, following Bauer and Swanson (2023a,b). This approach relies on interest rate surprises only, it does not use the stock price surprises and disregards the existence of the Central Bank Information effect. As neither the Monetary Policy shock nor the Fed Response to News shock is able to generate a positive co-movement between interst rates and stock prices, this approach does not account for the cases positive co-movement present in Figure 1, implicitly treating it as measurement error or noise. As we show next, events with positive co-movement are not the result of noise or measurement error either, because extra information inherent in the high-frequency stock-price surprises have significant impact on the evolution of the macroeconomy.

Figure 5 implements the "pure" Bauer-Swanson approach. In this approach, we assume that the Bauer-Swanson residual is a perfectly measured monetary policy shock, the Bauer-Swanson fitted value is a perfectly measured FRN shock, and we abstract from the possibility of a CBI shock. To track the effects of these shocks on the macroeconomy we identify the VAR recursively, putting the residual first, the fitted value second. We also include the S&P500 surprise as a third variable. To justify this approach, note that first, since the residuals and the fitted values are mutually orthogonal, their ordering does not matter, we only put the residual first to have the monetary policy effects in the first column as the preceding figures. Second, the inclusion of the S&P500 surprise as the third variable does not affect the estimated effects of MP and FRN shocks.

The results in Figure 5 are consistent with Bauer and Swanson (2023a) but at the same time raise a question about the comprehensiveness of their approach.

They are consistent, because, as they point out, controlling for the FRN effect captured by FIT reduces the attenuation bias in the interest rate surprises. The effects of RES shocks on the economy are significantly contractionary and consistent with theoretical predictions. The effects of FRN shocks are expansionary, indicating that they capture the attenuation bias.

However, the results in the third column suggest that something important is missing in the "pure" Bauer-Swanson approach. By including the S&P500 surprise after RES and FIT we remove all stock price surprise variation explained by the interest rate surprises (both "predictable" and "unpredictable" in sample), leaving only the

unexplained residual variation. One might think that the primary information in Fed communications is about interest rates and that stock price reaction reflect only this information, plus some noise. However, the results in Figure 5 challenge this view. It shows that stock-price movements not explained by interest-rate news are followed by highly significant responses of the key macroeconomic and financial variables: persistent increases in Treasury yields and stock prices, a decrease in the excess bond premium, and gradual rises in real GDP and its deflator. The lesson is that the high-frequency stock price response to Fed communication helps understand the consequences of the communication over and above the interest rate surprise. The point that stock price surprises carry important announcement-related information not spanned by the interest rate surprises is made also in Kroencke et al. (2021); Boehm and Kroner (2024).

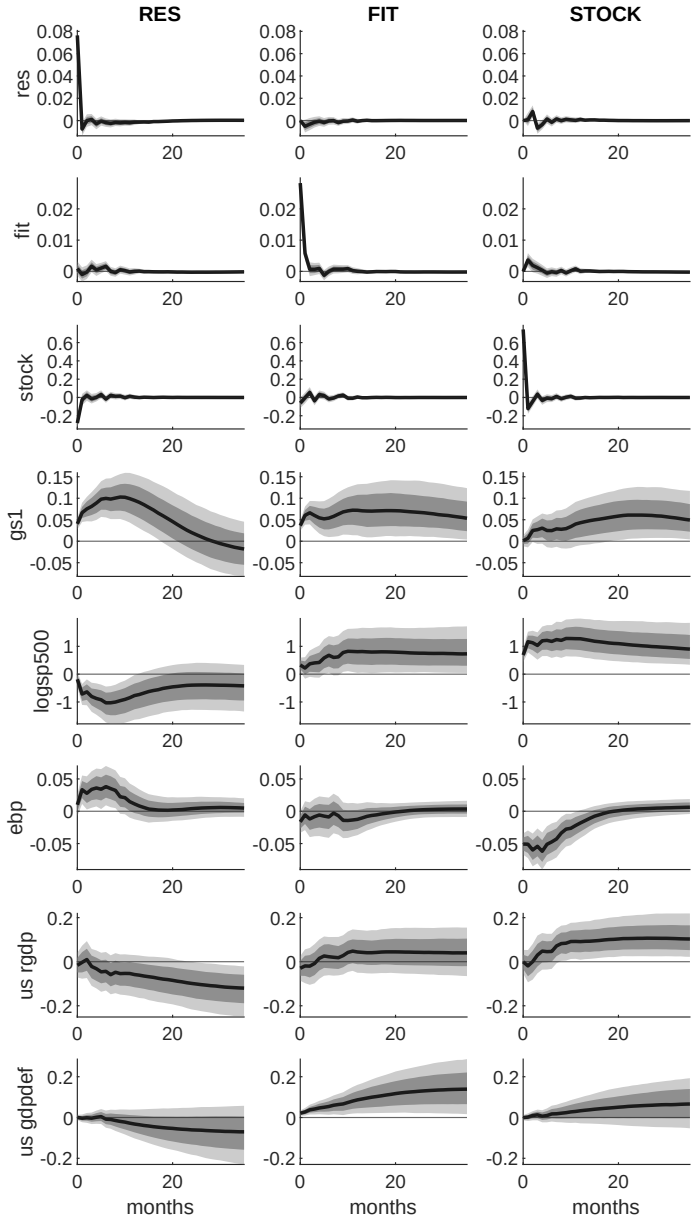
How important is it to control for both central bank information and Fed response to news effects when identifying monetary policy shocks? Figure 6 compares impulse responses to monetary policy shocks under different identification schemes. Our baseline methodology (Section 4.4) controls for both CBI and FRN effects, the FRN-only approach follows Bauer and Swanson (2023a), the CBI-only approach is based on Jarociński and Karadi (2020), with sharper inference using the heteroskedasticity of shocks between FOMC and non-FOMC events, and the fourth, naive approach uses only the high-frequency interest rate surprise as instrument. The impulse responses are broadly similar, but both the FRN-only and the CBI-only approaches eliminate some attenuation bias relative to the naive approach and imply a quantitatively larger impact on activity, prices and financial conditions with a somewhat smaller interest rate shock. Controlling for both channels leads to an even stronger effect (including a significant impact on the price level that the other methods fail to deliver) at an even smaller interest rate shock. This result indicates that controlling for both channels is important to arrive at unbiased estimates.

5. Identification using only sign and magnitude restrictions

In this section, we identify the shocks using only sign and magnitude restrictions outlined in Section 4.1, without bringing in the information contained in the heteroskedasticity of the data. We find that the point estimates of the effects on the economy are similar to the baseline results and are consistent with similar conclusions: both FRN and CBI effects bias the effects of monetary policy, with the CBI accounting for most of the bias in the monthly VAR. However, the error bands become too wide to make any precise statements about the relative role of FRN vs CBI shocks. This problem is further aggravated when we discard the information in the non-FOMC events and restrict ourselves to the FOMC sample.

When using sign restrictions alone, the set of possible identifications consistent with the sign and magnitude restrictions alone includes many different cases (Wolf

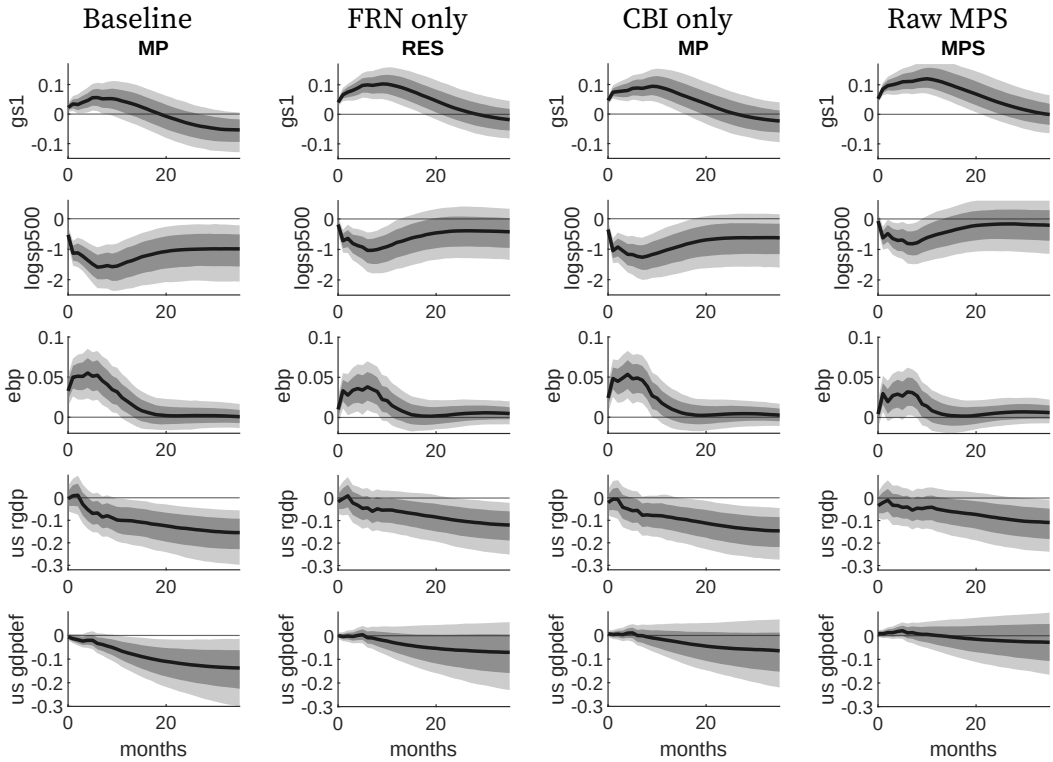
FIGURE 5. The "pure" Bauer-Swanson approach: dynamic effects of Bauer-Swanson residuals (RES), fitted values (FIT), and S&P500 surprises (SP500HF) in a monthly VAR. Baseline sample.



The results show that RES shocks have contractionary effects consistent with theoretical predictions, while FRN shocks are expansionary, indicating they capture an attenuation bias in interest rate surprises. The inclusion of SP500HF isolates stock price movements unexplained by interest rate news, revealing significant responses in key macroeconomic and financial variables omitted in an FRN-only approach. The two shades of grey show 68% and 90% posterior bands. The black solid line shows the median impulse response. The VAR is monthly 1988:01-2025:03. Baseline set of Fed events (both FOMC and non-FOMC).

2022). To give an example, we first characterize just one case: the median target shock (Fry and Pagan 2011). Table 5 reports the variance decomposition for this shock. Compared with the heteroskedasticity-informed identification, the Fed response to news (FRN) shock plays an even smaller role in explaining the Bauer-Swanson residual (RES) and the monetary policy shock (MP) plays an even smaller role in explaining the Bauer-

FIGURE 6. Impulse responses to monetary policy shocks across different identification schemes. Baseline sample.



Impulse responses to monetary policy shocks under the baseline (Section 4.4), under the FRN-only approach (Section 4.5), under a CBI-only approach (Section 4.5) and using the raw monetary policy surprise. The shaded areas show robust 68% and 90% bands. The black solid line shows the median impulse response. The VAR is monthly 1988:01-2025:03. Baseline set of Fed events (both FOMC and non-FOMC).

TABLE 5. Variance decompositions in the median rotation.

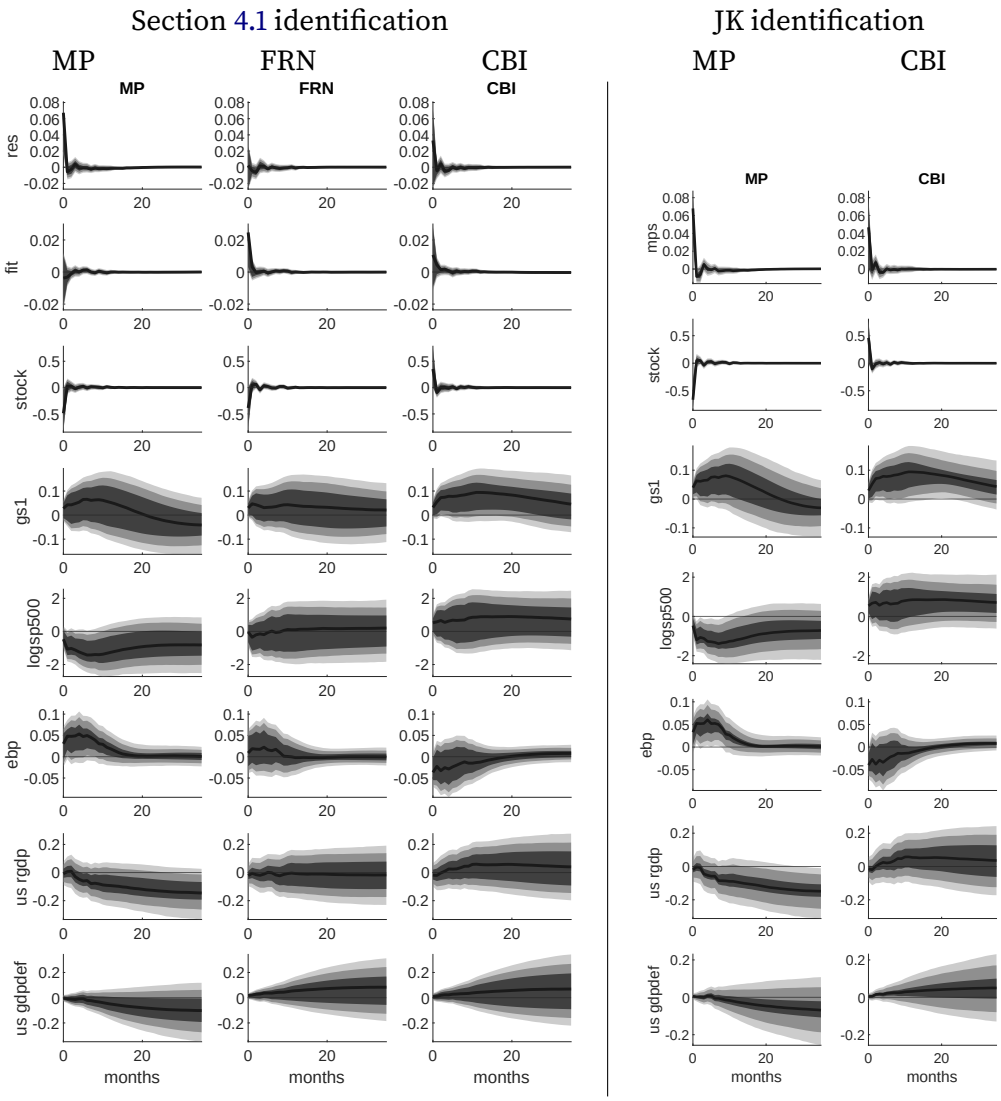
	res	fit	stock
MP contrib.	0.79	0.04	0.46
FRN contrib.	0.00	0.83	0.22
CBI contrib.	0.21	0.13	0.32

Swanson fitted value (FIT). Like in the heteroskedasticity-informed case, the interest rate surprises are affected by the CBI shock. Unlike in the heteroskedasticity-informed case, where the CBI shock affects mainly the RES, in the median rotation central bank information (CBI) shock also explains a nontrivial fraction of FIT: the CBI accounts for 21% of the variance of RES and for 13% of the variance of FIT.

We report the dynamic effects of the median shocks in a daily local projection in the Appendix Figure A3. The results are similar to the results reported in Figure 3.

We next study the effects of sign-restriction identified shocks in a monthly VAR. The VAR includes the same low-frequency variables as before, but a different set of high-frequency variables. The high-frequency variables are now the residual (res)

FIGURE 7. Dynamic effects of MP, FRN and CBI shocks identified by sign and magnitude restrictions only. Monthly VAR with the robust bands of Giacomini and Kitagawa (2021). Baseline sample.



While the median rotations are broadly similar to the baseline, the results illustrate the challenges with obtaining precise estimates with only sign and magnitude restrictions. The darkest region shows the set of posterior means. The two lighter shades of grey show robust 68% and 90% bands. The black solid line shows the median impulse response. The VAR is monthly 1988:01-2025:03. Baseline set of Fed events (both FOMC and non-FOMC).

and fitted value (fit) of the Bauer-Swanson regressions described in Section 3.3, and surprises in the S&P500 stock market index calculated over our baseline Fed events sample including both FOMC and non-FOMC events. The high-frequency variables are aggregated to the monthly frequency by adding them up. In this VAR we identify three shocks using the sign and magnitude restrictions from Section 4.1.

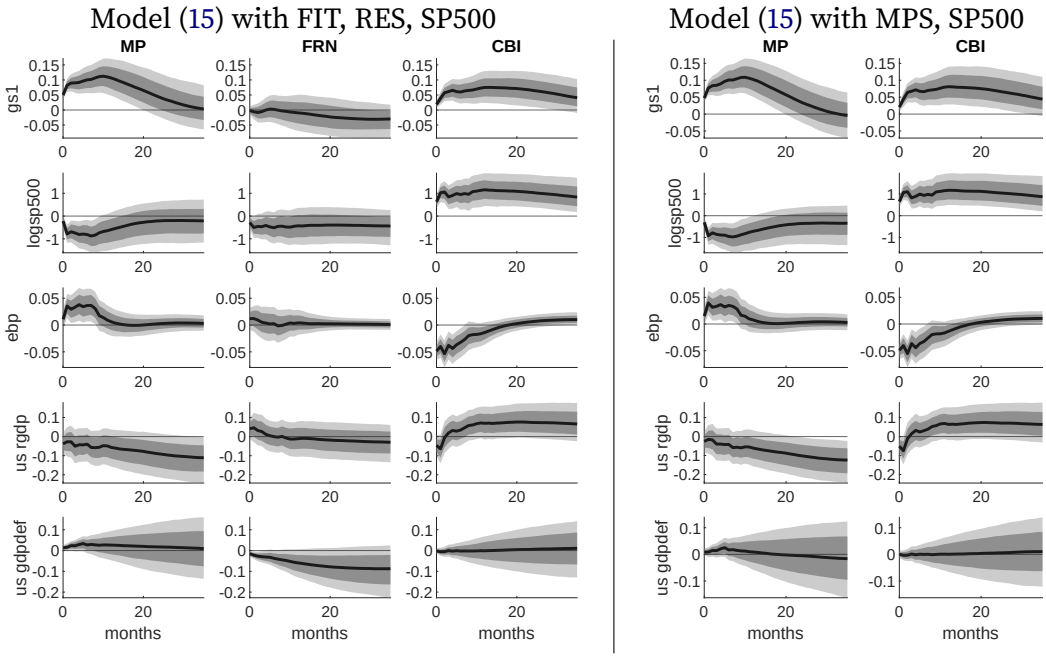
Figure 7 reports the full range of impulse responses identified by our sign restrictions, obtained with the methodology of Giacomini and Kitagawa (2021). The dark regions show the full set of posterior means consistent with sign restrictions, while

the shades of grey show robust 68% and 90% posterior bands. Monetary policy shocks are contractionary for virtually all rotations consistent with the sign restrictions. For CBI and FRN shocks the sets of posterior means overlap to a larger extent and include zeros, so it is harder to draw strong conclusions about their relative roles.

According to the posterior median impulse response, plotted with the solid line, the CBI shock is more expansionary than the FRN shock, i.e. it generates a more severe attenuation bias when not accounted for. The impulse responses of the median FRN shock are close to zero, implying that its attenuation bias is close to zero, in line with our baseline result.

In the Appendix we report the sign-restrictions-only VARs identified using only the FOMC events. In this case, of course, as we only have a single subsample, we have no option of using heteroskedasticity and we can only rely on the sign restrictions. The results, reported in Figure A4, are, if anything, even less informative: the posterior regions are wide and it is hard to judge the relative importance of the FRN and CBI contamination with any confidence.

FIGURE 8. Dynamic effects of MP, FRN and CBI shocks in a monthly VAR. Heteroskedasticity-based identification. All FOMC and all non-FOMC events.



6. Robustness

Using all non-FOMC events (not just the "big" ones). We repeat the analysis using all non-FOMC events, instead of only the big non-FOMC events. Figure A5 reports the scatterplots of interest rate and stock price surprises across the two subsamples. Heteroskedasticity-based identification applied to these data yields similar MP and CBI shocks as before, but no shock that looks like the FRN shock. More in detail, heteroskedasticity in this case pins down a shock that raises interest rates, predominantly the Bauer-Swanson residual, with contractionary effects on the financial markets and the economy (MP shock). It also pins down a shock that raises interest rates, again predominantly the Bauer-Swanson residual, with expansionary effects on the financial markets and the economy (CBI shock). However, the third shock, which would be a candidate for the FRN shock, hardly moves the interest rates and has unclear effects on financial markets and the economy. Moreover, neither the MP nor the CBI shock has much effect on the price level. Our interpretation is that by using all non-FOMC events we introduce too much noise in the system and our identification approach fails.

Using subsets of the Bauer-Swanson regressors. In this exercise, when running the Bauer-Swanson regressions we only include the explanatory variables used in Bauer and Swanson (2023b).

TABLE 6. Bauer-Swanson regressions over the baseline, the FOMC and the non-FOMC samples: AER regressors only

	Baseline	FOMC	Non-FOMC
(Intercept)	-0.011	-0.028***	0.006
	(0.008)	(0.010)	(0.013)
Unemployment surprise	0.016	0.013	0.013
	(0.017)	(0.020)	(0.028)
Payrolls surprise	0.012	0.011	0.021
	(0.012)	(0.013)	(0.040)
Real GDP surprise	0.012**	0.003	0.019**
	(0.006)	(0.007)	(0.009)
Core CPI surprise	0.033	0.026	0.010
	(0.026)	(0.033)	(0.041)
Core CPI median forecast	0.015	0.064*	-0.031
	(0.029)	(0.036)	(0.046)
6-month change in core CPI	-0.003	0.001	-0.006
	(0.004)	(0.005)	(0.007)
BBK index	0.000	0.001	-0.001
	(0.001)	(0.002)	(0.002)
$\Delta \log$ S&P500 (3m)	0.048	0.128**	-0.013
	(0.040)	(0.055)	(0.057)
Δyield curve slope (3m)	-0.019	-0.077***	0.029
	(0.023)	(0.030)	(0.035)
$\Delta \log$ pcommodity (3m)	0.094**	0.164***	0.021
	(0.037)	(0.050)	(0.054)
N	696	357	339
R-squared	0.029	0.120	0.028
Adj.R-squared	0.015	0.095	-0.001

7. Conclusion

This paper develops a methodology to separately identify monetary policy (MP) shocks, central bank information (CBI) shocks, and Fed response to news (FRN) shocks and extends the sample of FOMC statements with non-FOMC Fed announcements, and. To achieve this, it uses (i) the high-frequency co-movement of interest rates and stock prices, (ii) the predictability of surprises by fresh public news, and (iii) the heteroskedasticity between FOMC and non-FOMC events. Our findings confirm the significant presence of CBI effects and argues that insufficiently controlling for them would lead to biased monetary policy shock estimates. The Fed response to news channel is significant in daily local projections, but it has a marginal impact in our baseline monthly VAR specification. Monetary policy shocks, when purged of CBI and FRN effects, generate macroeconomic responses consistent with theoretical expectations, including a temporary downturn in economic activity and prices and a deterioration in financial conditions.

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Appendix A. Structural Model

This section assesses whether a standard New Keynesian model can rationalize the empirical findings under different monetary policy shock identifications. Specifically, we ask: what combination of nominal and financial frictions is required to match observed responses to (i) our baseline monetary policy shock (purged of FRN and central bank information effects) versus (ii) the standard high-frequency identification and (iii) when only either the FRN or the CBI effects are purged from the measure? This exercise serves as a consistency check on our identification strategy through the lens of a structural model.

We employ a widely-used New Keynesian framework with balance sheet constrained financial intermediaries, closely following Gertler and Karadi (2011). The framework is well suited to analyze the quantitative impact of monetary policy shocks, which are modeled as temporary deviations from a systematic interest rate rule. In the model, monetary policy influences real allocations through two key frictions: nominal rigidities and financial frictions..

We structure the description of the model below along the lines of the transmission of monetary policy. To conserve space, we describe key equilibrium conditions and refer the reader to the original paper for their derivations. The framework has 7 agents: representative households, financial intermediaries, intermediate-good and capital-good producers, retailers, a fiscal authority, and a central bank. Representative households consume a basket of differentiated goods, work, and save. Financial intermediaries collect deposits and lend to intermediate good firms. Intermediate good firms use capital and labor to produce intermediate goods and borrow from financial intermediaries and households to finance capital acquisitions. Capital-good producers use final goods to generate new capital. Retailers purchase intermediate goods, differentiate them, and sell them to households. The fiscal authority finances its exogenous expenditures with lump sum taxes. The central bank sets interest rates.

A.1. Central bank

The central bank sets the nominal interest rate (i_t) following a Taylor rule:

$$i_t = \kappa_\pi \pi_t + \kappa_x x_t + \varepsilon_t, \tag{A1}$$

where π_t denotes the inflation rate, x_t is a measure of economic slack, $\kappa_\pi > 1$ and $\kappa_x > 0$ are parameters. We proxy economic slack with the log deviation of marginal cost of the intermediate good from its steady state, which is proportional to conventional output gap measures. The policy temporarily deviates from its systematic component due to monetary policy shocks (ε_t). The shock follows a second-order autoregressive process $\varepsilon_t = \rho_1^{MP}\varepsilon_{t-1} + \rho_2^{MP}\varepsilon_{t-2} + \epsilon_t^{MP}$, where we allow for AR(2) dynamics to better capture the hump-shaped response of the monetary policy indicators observed in the data.

A.2. Nominal rigidities

The real interest rate (r_t) is determined by the Fisher equation

$$i_t = r_t + E_t \pi_{t+1}. \tag{A2}$$

Monetary policy influences the real rates temporarily as a result of nominal rigidities. Nominal wages are flexible; nominal rigidities arise from staggered price setting by retailers. Their behavior implies a standard New Keynesian Phillips curve with a backward-looking term:

$$\pi_t - \gamma_P \pi_{t-1} = \beta \left(E_t \left\{ \pi_{t+1} \right\} - \gamma_P \pi_t \right) + \frac{(1-\gamma)(1-\beta\gamma)}{\gamma} x_t, \tag{A3}$$

where β is the steady state discount factor of the representative household, $\gamma \in [0,1]$ is the Calvo parameter (the probability of unchanged prices), and $\gamma_P \in [0,1]$ is the share of prices indexed to previous period inflation. These two parameters jointly determine price rigidity. The Calvo parameter determines the sensitivity of inflation to marginal cost (x_t): a high γ translates into low sensitivity, implying that the price level responds sluggishly to monetary policy disturbances that change marginal costs. Indexation influences how backward-looking the relationship is, with high γ_P implying high persistence in the inflation rate.

A.3. Real effects of monetary policy

The real interest rate influences aggregate demand through its impact on consumption, investment, and indirectly on government expenditures. Consumption is governed by the representative household's Euler equation:

$$E_t \left\{ \Lambda_{t,t+1} R_{t+1} \right\} = 1, \tag{A4}$$

where $R_t = \exp(r_t)$ is the gross real interest rate, and $\Lambda_{t,t+1}$ is the stochastic discount factor, given by

$$\Lambda_{t,t+1} = \beta \frac{\rho_{t+1}}{\rho_t}, \tag{A5}$$

where ρ_t is the marginal utility of consumption:

$$\rho_t = (C_t - hC_{t-1})^{-1} - \beta h E_t (C_{t+1} - hC_t)^{-1}, \tag{A6}$$

with $h \in [0,1]$ governing the strength of consumption habits.

A persistent increase in the real rate following a monetary policy shock raises the opportunity cost of current consumption relative to future consumption. This reduces consumption, and the impulse response takes an empirically realistic hump-shaped form due to habits.

Investment is determined by capital good producers, who transform consumption

goods into capital goods subject to an investment adjustment cost function (f) and sell them to intermediate good firms at price Q_t:

$$Q_t = 1 + f\left(\frac{I_t}{I_{t-1}}\right) + \frac{I_t}{I_{t-1}} f'\left(\frac{I_t}{I_{t-1}}\right) - E_t \Lambda_{t,t+1} \left(\frac{I_{t+1}}{I_t}\right)^2 f'\left(\frac{I_{t+1}}{I_t}\right) \tag{A7}$$

An increase in real rates reduces the value of capital Q_t, which equals the present discounted value of future capital returns. It declines because, first, higher real rates cause a downturn and reduce the marginal product value of capital and, second, higher interest rates increase the discount rate applied to future dividends. Low capital prices reduce incentives to invest, generating a realistic hump-shaped decline in investment due to the functional form of adjustment costs. Aggregate capital (K_{t+1}) evolves according to: $K_{t+1} = I_t + (1-\delta)K_t$.

Government expenditure is assumed to be a fraction of aggregate output $G_t = \exp(\bar{g})Y_t$, where $\bar{g}$ is a constant. Aggregate demand net of investment adjustment costs equals the sum of consumption, investment and government expenditures.

Aggregate demand is fulfilled through the supply of intermediate good producers that serve the retailers. Intermediate goods producers combine capital and labor in a constant returns to scale technology:

$$Y_{mt} = AK_t^{\alpha} L_t^{1-\alpha}, \tag{A8}$$

where Y_{mt} is intermediate good production, $A = \exp(a)$ is a measure of aggregate technology (constant for simplicity), L_t is labor, and α is the capital income share. We denote the price of the intermediate good P_{mt}. The marginal product value of capital is $Z_t = P_{mt}\alpha\frac{Y_t}{K_t}$. Equilibrium in the labor market requires $P_{mt}(1-\alpha)\frac{Y_{mt}}{L_t} = \chi\rho_t^{-1}L_t^{\varphi}$, where χ is the relative utility weight of leisure and φ is the inverse Frisch elasticity of labor supply.

A.4. Financial frictions

We now describe how financial frictions are introduced into the model. Intermediate-good firms issue state-contingent corporate bonds S_t that they use to finance purchases of capital (K_{t+1}) from capital producers. They supply corporate bonds at the value

$$Q_t S_t = Q_t K_{t+1}, \tag{A9}$$

where Q_t is the real value of capital. The corporate bonds pay the marginal product value of capital (Z_t) every period and decay geometrically with a parameter $1-\delta$, where δ is capital depreciation rate. Therefore, their value (Q_t) equals to the value of the

capital.[10] The (gross) corporate bond return is

$$R_{kt} = \frac{Z_t + (1-\delta)Q_t}{Q_{t-1}}. \tag{A10}$$

Demand for corporate bonds comes from both financial intermediaries (or bankers) and households:

$$S_t = S_{bt} + S_{ht}. \tag{A11}$$

Bankers are part of a household with perfect consumption insurance. They continue as a banker each period with probability $\sigma \in [0,1]$, and exit and return their net worth to the household with the complementary probability $1-\sigma$. The share of bankers is kept constant by assuming that some workers become bankers every period. New bankers receive startup funds from the households. The aggregate startup funds amount to ω. Banks collect deposits from households and pay them the gross real return R_t. They combine deposits with their net worth and invest them into corporate bonds.

Financial intermediaries face an agency friction: they can abscond with a fixed fraction of assets under their management, losing the franchise value of their banking license. To prevent this, households limit deposits placed in financial intermediaries, effectively setting an endogenous leverage (ϕ_t) constraint. The leverage constraint ensures banks have enough 'skin in the game' to avoid absconding. It limits the amount of corporate lending financial intermediaries can supply (S_{bt}):

$$Q_t S_{bt} = \phi_t N_t, \tag{A12}$$

where N_t is the aggregate net worth of the banking system.

Amplification mechanism.. How do financial frictions amplify the impact of a monetary policy shock? A temporary increase in the nominal rate translates into a higher real rate r_t due to nominal rigidities. Higher real rates reduce consumption through a standard intertemporal substitution mechanism. Furthermore, higher real rates raise the funding costs of banks, causing them to require higher returns on corporate bonds ($E_t R_{kt+1}$). The higher discount rate on existing capital reduces its value Q_t, lowering incentives for investment. This channel operates even without financial frictions (with lax bank balance sheet constraints).

Binding leverage constraints amplify the shock through standard financial accelerator mechanisms. Lower corporate debt values reduce banking sector assets, deteriorating their balance sheet condition. The asset price drop causes an amplified decline in net worth, with a multiplicative factor equal to their leverage. Deteriorating balance sheet conditions further increase credit costs and worsen credit conditions,

[10] The corporate bonds can be understood as equity. Firms operate a constant returns to scale technology without profits. So the value of the firm comes only from the value of their capital holdings.

with additional negative impacts on investment. The worsening outlook further reduces asset prices, creating another negative feedback loop.

The financial intermediaries build net worth from retained earnings and from start-up funds. Aggregate net worth evolves according to the following law of motion:

$$N_t = \sigma\left[(R_{kt} - R_t)\phi_{t-1} + R_t\right]N_{t-1} + \omega. \tag{A13}$$

The first term captures net worth from retained earnings of surviving bankers, while the second term comes from start-up funds of new bankers. Retained earnings are scaled by the survival probability σ because exiting bankers repay their net worth as dividends. Surviving bankers' retained earnings come from two terms: banks earn the gross real return R_t on their net worth and an excess return $R_{kt} - R_t$ on corporate bond holdings, the latter equaling the product of their net worth and leverage ϕ_{t-1}.

Households also lend directly to the corporate sector, subject to a portfolio adjustment cost as in Gertler and Karadi (2013). The household must pay $\kappa(S_{ht} - \bar{S}_h)^2$ if it purchases corporate bonds in excess of $\bar{S}_h$, where $\kappa \geq 0$ is a portfolio adjustment cost parameter. This parameter serves as our key measure of financial friction severity: as $\kappa \to 0$, households fully offset constrained bank lending, effectively eliminating financial frictions from the model. The household demand for corporate bonds is determined by

$$S_{ht} = \bar{S}_h + \frac{1}{\kappa}E_t\Lambda_{t,t+1}\left(R_{kt+1} - R_{t+1}\right), \tag{A14}$$

where $\Lambda_{t,t+1}$ is the household's stochastic discount factor. The demand function implies that households respond to increases in corporate bond spreads by increasing their holdings. The parameter κ determines the sensitivity of their response. Importantly, as $\kappa \to 0$, households increase holdings without limits for any positive premium, issuing credit to intermediate good firms without constraints and fully replacing the constrained banking sector.

A.5. Pricing additional assets

Our baseline VAR includes a 1-year government bond yield and the excess bond premium. The latter is a yield spread between corporate and government bonds with an average duration of around 7 years. In order to obtain analogous long-term yields in our model, we introduce multiple long-term bonds as perpetuities with geometrically decaying coupons. We calibrate the rate of decay of their coupons (σ_x) to match their duration. The assets are priced through no-arbitrage conditions but are not held in positive quantities in equilibrium. Government bonds are priced by households, who are assumed to trade them without portfolio adjustment costs. Corporate bonds, by contrast, are traded by the banks, which require excess return.

We denote by q_{xt} the nominal price of a government bond with duration x. It pays σ_x^i unit in each quarter $i = 0, 1, 2, \ldots$. Its steady state (yearly) duration is $1/[4(1-\sigma_x/R)]$, where R is the steady state gross real rate (and steady state inflation is 0). Its (gross)

nominal yield to maturity is $Y_{xt} = 1/q_{xt} + \sigma_x$. The no arbitrage condition requires:

$$R_{t+1}\Pi_{t+1} = \frac{1 + \sigma_x q_{xt+1}}{q_{xt}}. \tag{A15}$$

Analogously, we denote by Q_{xt} the nominal price of a corporate bond with duration x. It pays σ_{kx}^i units in periods $i = 0, 1, 2, \ldots$. Its steady state duration is $1/[4(1 - \sigma_{kx}/R_k)]$, where R_k is the steady state corporate bond return. Its gross yield to maturity is $Y_{kxt} = 1/Q_{xt} + \sigma_{kx}$. The no arbitrage condition implies:

$$R_{kt+1}\Pi_{t+1} = \frac{1 + \sigma_{kx} Q_{xt+1}}{Q_{xt}}. \tag{A16}$$

The (gross) excess bond premium in our model is measured as $EBP_t = Y_{kxt}/Y_{xt}$.

A.6. Calibration

We solve the model through first-order perturbation around a non-stochastic steady state. We estimate key parameters of the model through a standard impulse response matching exercise (Christiano, Eichenbaum and Evans 2005). In particular, we estimate three parameters: (i) the Calvo parameter γ, (ii) the inflation indexation parameter γ_P and (iii) the household portfolio adjustment cost parameter κ, together with the size and persistence of the monetary policy shock (σ_{MP}, ρ_1^{MP}, ρ_2^{MP}), to match impulse responses to a monetary policy shock in the model and in the VAR. The first two parameters determine the level of nominal frictions, and the third parameter influences the level of financial frictions in the model. Other model parameters are standard and borrowed from Gertler and Karadi (2011).

Our baseline impulse responses include 5 variables: the 1-year government bond yield, the GDP and the GDP deflator, the S&P500 stock market index, and the excess bond premium. In the model, we match these with the deviations of the following 5 variables from their steady state values: yield to maturity of a 1-year government bond ($\hat{y}_{1t}$), the output $\hat{y}_t$, the price level $\hat{p}_t = \sum_{s=1}^{t} \hat{\pi}_s$, the value of capital Q_t ($\hat{q}_t$)[11] and the excess bond premium ($\hat{ebp}_t$).

We transform monthly VAR impulse responses into quarterly impulse responses by taking simple averages over each quarter, giving us 12 moments for each observables. We simulate impulse responses from the model and stack the 5 times 12 differences of the VAR and model moments into a vector V. We estimate our model parameters to minimize $V'\Omega V$ scalar, where Ω is a weighting matrix. Following Christiano, Eichenbaum and Evans (2005), Ω contains the diagonal elements of the inverse of the variance-covariance matrix of the moments from the VAR.

[11] The equity value of financial intermediaries (N_t) in the model would be an alternative model variable for the S&P500. The two variables move in tandem in the model, but the letter gets amplified by the leverage, as S&P500 valuations are amplified by the average leverage of the financial and non-financial firms. We report both the evolution of Q and N on our figures and our results are robust to using N_t as a measure of stock market valuations.

FIGURE A1. Matched impulse responses to monetary policy shock, baseline and standard high-frequency identification, Model (black line), VAR mean (blue dashed line), 2-standard-deviations band.

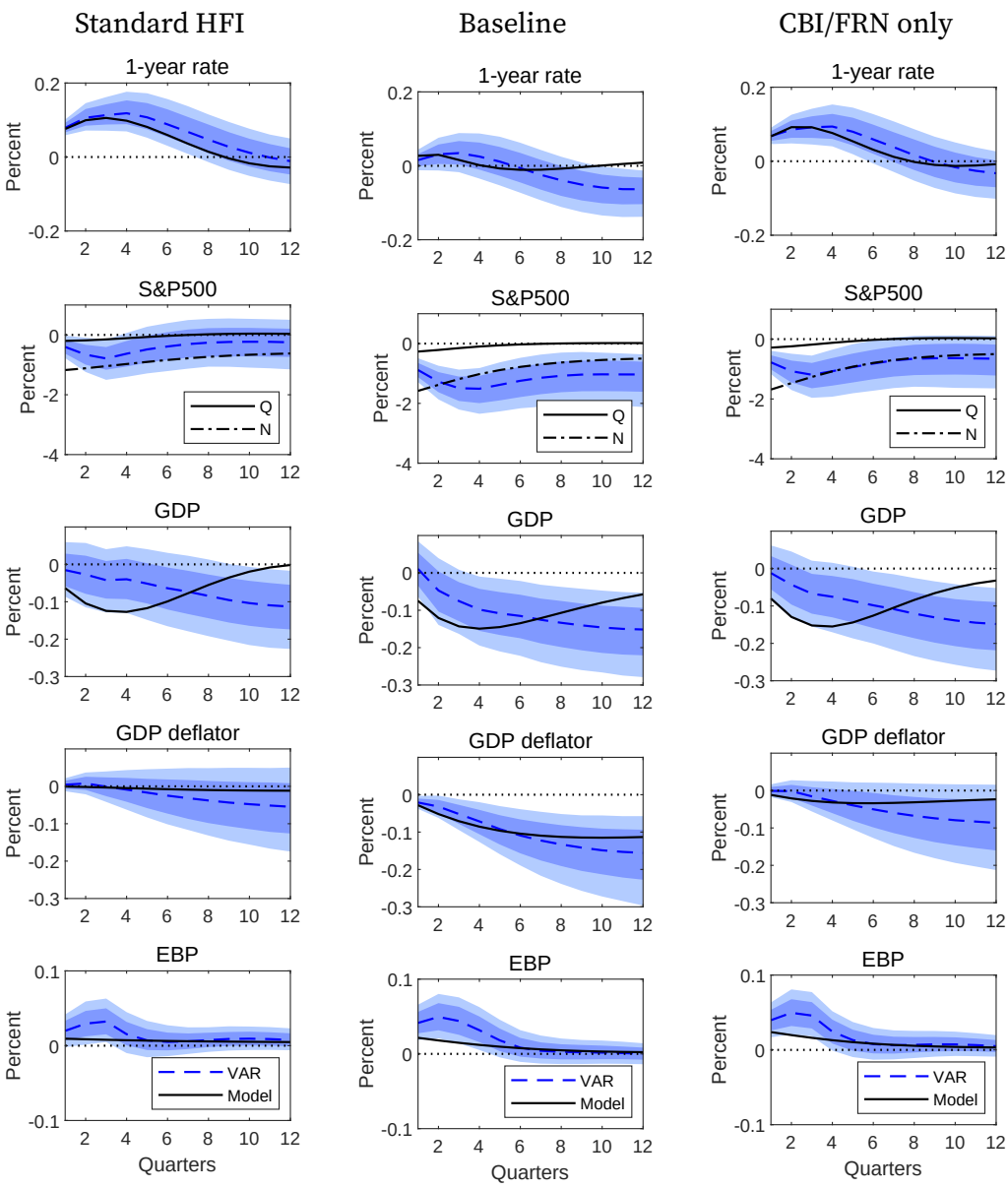


Table A1 lists the estimated parameter values. Figure A1 shows the impulse responses implied by the model and compares them to the impulse responses from the VAR.

Standard high-frequency identification. We first conduct the exercise using the impulse responses to the monetary policy shock from the standard high-frequency identification, which disregards both Fed response to news effects and central bank information shocks. The first column of Table A1 and Figure A1 show the results. The price level

TABLE A1. Estimated parameters

Parameters	Label	Standard HFI	Baseline	CBI/FRN only
Calvo parameter	γ	0.98	0.87	0.91
Inflation indexation	γ_P	0.996	0.00	0.00
Portfolio adjustment cost	κ	0.01	0.055	0.071
Stdev of monpol shock	σ^{MP}	0.0003	0.0006	0.0004
AR(1) parameter of monpol shock	ρ_1^{MP}	1.63	1.14	1.48
AR(2) parameter of monpol shock	ρ_2^{MP}	-0.74	-0.33	-0.63

response is unreasonably sticky in this case, and the model requires extreme price-stickiness and indexation parameters to capture the impact. These parameters imply that prices are reset on average every 4 years, far exceeding the 6–12 month frequencies documented in microdata studies (**??**). With such a high nominal stickiness, the interest rate shock causes an output decline that severely overestimates the responses predicted by the VAR, especially in the early years. This occurs even though the size and the persistence of the monetary policy shock somewhat underestimates the observed yield responses. Relatedly, financial frictions are estimated to be tiny: the model predicts close to zero corporate bond spread response, inconsistently with the VAR evidence. Had it estimated higher financial amplification, the model would have fared even worse in matching the observed output response.

Baseline identification. Next, we conduct the exercise using our baseline identification. This monetary policy shock is purged from the impact of both the central bank information shock and the Fed response to news effects. The second column of Table A1 and Figure A1 show the results. The monetary policy shock matches well the observed yield response. The price stickiness is now estimated to be smaller and the model requires no backward indexation to match the observed price level response. The Calvo parameter implies prices are reset somewhat more frequently than once every two years, which is still higher than microdata evidence but not unreasonable given that our simple model does not account for wage stickiness. The more moderate price stickiness, however, is insufficient to explain the output response, so the model estimates a sizable financial friction parameter—an order of magnitude larger than in the standard high-frequency identification. This approach also better matches the observed reaction of the excess bond premium.

One might wonder how small yield movements in our baseline calibration generate sizable output and price responses. The answer lies in general equilibrium feedback through the Taylor rule.

The underlying monetary policy shock raises the federal funds rate by approximately 25bps (not shown). This generates declines in inflation (>10bps) and output gap

(>15bps). However, the Taylor rule responds contemporaneously to these deteriorating conditions by easing policy, partially offsetting the initial shock. The net equilibrium yield response peaks at only 5 basis points.

Controlling for CBI or FRN only.. Lastly, we conduct the exercise using the monetary policy identification controlling for either only the Fed response to news effect or the central bank information effect. The two estimates are sufficiently close that they yield similar parameter estimates. The third column of Table A1 and Figure A1 show the results. Controlling for only one of the effects goes a long way in generating more reasonable nominal stickiness parameters: the model needs no backward indexation and its nominal stickiness parameter is more reasonable than under the standard HFI. Still, the magnitude of required nominal stickiness remains unreasonably high and significantly higher than under our baseline parametrization.

Implications.. These results demonstrate that our baseline identification yields model implications broadly consistent with both microeconomic price-setting evidence and significant financial frictions. By contrast, the standard high-frequency identification requires implausibly rigid prices and minimal financial frictions—a combination inconsistent with both microdata and the observed credit spread responses. Therefore, our baseline identification approach is in line with a structurally more plausible model.

Appendix B. Construction of high-frequency surprises

Table A2 reports the correlations of the selected variables in the GSS dataset with the corresponding variables in our dataset. The GSS dataset we are comparing with includes only FOMC announcements, and has data up to July 2019. The correlations are computed on the common samples of events, which may differ by variable. The last column reports the size of the common sample for each variable. The correlations are very high, between 0.938 and 0.989, showing that in spite of the differences in the computation and data sources we replicate the GSS dataset very closely.

TABLE A2. Correlations of selected variables in the GSS dataset (2019 version) and in our dataset

Variable	GSS identifier	JK identifier	Correlation	N.obs.
Fourth Fed funds future	FF4	FF4	0.959	195
Fed funds rate after the next FOMC	MP1	MP1	0.988	197
Front Eurodollar future	ED1	ED1	0.979	285
First Back Eurodollar future	ED2	ED2	0.988	285
Second back Eurodollar future	ED3	ED3	0.989	285
Third back Eurodollar future	ED4	ED4	0.925	285
2-year Treasury yield	ONRUN2	TFUT02	0.958	242
5-year Treasury yield	ONRUN5	TFUT05	0.981	242
10-year Treasury yield	ONRUN10	TFUT10	0.966	242
30-year Treasury yield	ONRUN30	TFUT30	0.941	242
S&P500 (spot)	SP500	SP500	0.938	284
S&P500 (futures)	SP500FUT	SP500FUT	0.971	274

Table A3 provides information on the adjustment of the Fed event windows to account for interfering events.

TABLE A3. Interfering events

Interfering Event \ Fed Event Type	FOMC announcement	Chair speech	Testimony	Testimony release	Total
US Auction Announcement Note	0	23	89	0	112
U.S.: Composite Index of Leading Indicators	0	9	14	0	23
US Auction Announcement Bond	0	9	13	0	22
U.S.: New Home Sales	0	4	17	0	21
U.S.: University of Michigan Consumer Sentiment: Final	0	19	2	0	21
U.S.: Existing Home Sales	0	7	12	0	19
US Auction Announcement Note(Fed)	0	4	14	0	18
U.S.: ISM: Mfg Composite Index	0	6	9	0	15
U.S.: Chicago Purchasing Managers Index	0	6	7	0	13
U.S.: Initial Claims	0	11	2	0	13
U.S.: ISM: Services Index	0	3	5	0	8
U.S.: Consumer Price Index	0	6	1	0	7
U.S.: New Orders: Advance Durable Goods	0	3	1	1	5
U.S.: Employees on Nonfarm Payrolls	0	5	0	0	5
U.S.: Producer Price Index	0	2	0	2	4
U.S.: Retail Sales	0	4	0	0	4
U.S.: PCE	0	2	0	0	2
U.S.: Real GDP: Final	1	1	0	0	2
US Auction Announcement Bond(Fed)	0	0	1	0	1
US Auction Announcement TIPS	0	1	0	0	1
U.S.: Real GDP: Advance	0	1	0	0	1
Total	1	126	187	3	317
Total (unique events)	1	107	150	3	261

Post-FOMC press conferences and the Releases of FOMC meeting minutes never coincided with the Interfering Events. The remaining Interfering events, which we considered, but which never coincided with the Fed events, are: ECB Press Releases, U.S.: Business Inventories, U.S.: Capacity Utilization Rate: Total Industry, U.S.: Consumer Confidence, U.S.: Consumer Credit, U.S.: Core CPI, U.S.: Government Surplus/Deficit, U.S.: Housing Starts, U.S.: Industrial Production, U.S.: Personal Income, U.S.: Real GDP: Preliminary, U.S.: Unemployment Rate, U.S.: Univ of Michigan Consumer Sentiment: Preliminary.

Some Fed events coincided with more than one Interfering Events. The row "Total (unique events)" counts the events of each type whose windows were adjusted by at least one Interfering Event.

Appendix C. Bayesian estimation of the i.i.d. heteroskedastic model

The following model relates the vector of N observables y_t (here: high-frequency surprises around Fed events) to the underlying N structural (orthogonal) shocks u_t.

$$y_t = C'u_t, \qquad u_{nt} \sim \mathcal{N}(0, \sigma^2_{ns}), \quad \text{for} \quad n = 1, \ldots, N, \quad s = 1, \ldots, S \tag{A17}$$

$t = 1, \ldots, T$ indexes the observations. C collects the effects of the shocks u_t on the observable quantities y_t. The key feature of the model is that each observation t belongs to one of S regimes ($2 \leq S \leq T$). The variance of shock n in regime s is σ^2_{ns}. This variance is allowed to be different in each regime.

Matrix C is identified as long as there is sufficient variation in the σ^2_{ns} across regimes. In particular, the mix of shocks needs to be different across regimes, i.e. the variances of different shocks should not change by the same proportion. This insight is due to Rigobon (2003). This paper and the subsequent literature (e.g., Lanne et al. 2010; Lütkepohl and Woźniak 2020; Sims 2021) provide formal conditions and/or tests for identification. The estimation approach outlined below breaks down (it runs into numerical problems) if the heteroskedasticity in the sample is not sufficient to identify C.

Model (A17) is estimated with a Bayesian approach, using the Gibbs sampler with a Metropolis step. For closely related estimation approaches see Lütkepohl and Woźniak (2020); Brunnermeier et al. (2021). The approach below uses a different strategy of sampling the C matrix than either of these papers. With the prior structure used in this paper, the extreme case of $S = T$ regimes is equivalent to the model with Student-t shocks and the Gibbs sampler below is a variant of the Gibbs sampler in Jarociński (2024).

C.1. Reparameterization, priors

Define $W = C^{-1}$ and $q_{ns} = (\sigma^2_{ns})^{-1}$ and reparameterize (A17) as follows:

$$y_t = W^{-1\prime}u_t = W^{-1\prime}Q_t^{-1/2}z_t, \quad z_t \sim \mathcal{N}(0, I_N), \quad Q_t = \mathrm{diag}(q_{1t}, \ldots, q_{Nt}) \tag{A18}$$

where $q_{nt} = q_{ns}$ when $t \in s$.

The likelihood of y_t in this parameterization is

$$p(y_t|W, Q_t) = |W^{-1\prime}Q_t^{-1}W^{-1}|^{-1/2} \exp\left(-\frac{1}{2}y_t'(W^{-1\prime}Q_t^{-1}W^{-1})^{-1}y_t\right) \tag{A19}$$

The prior for W is a weakly informative Gaussian distribution,

$$p(W) = \mathcal{N}(\mathrm{vec}\, W_0, \Omega). \tag{A20}$$

We take $W_0 = I_N$ and $\Omega = \kappa I_{N^2}$ where I_k denotes the identity matrix of order k. κ is a number large enough to ensure that the prior plays essentially no role. In practice we

take $\kappa = 200$ as in Brunnermeier et al. (2021).

The prior for q_{ns} is a Gamma distribution with shape and scale both equal to $\nu/2$ for all n and s,

$$p(q_{ns}) = \mathcal{G}(\nu/2, 2/\nu) = \Gamma(\nu/2)^{-1}(\nu/2)^{\nu/2} q_{ns}^{\nu/2-1} \exp(-q_{ns}\nu/2). \tag{A21}$$

Note that we parameterize the Gamma distribution with shape a and scale b as

$$\mathcal{G}_x(a, b) = \frac{1}{\Gamma(a) b^a} x^{a-1} \exp(-x/b) \tag{A22}$$

with mean ab and variance ab^2. We take $\nu = 10$ implying that the prior has the mean of 1 and variance 0.2. This prior distribution is conservative in the sense of acting against finding any heteroskedasticity and instead shrinking all the variances to the same value of 1.

Let $Y = (y_1, \ldots, y_T)'$ be the matrix collecting the observations of y_t and let $U = (u_1, \ldots, u_T)'$ be the matrix collecting the shocks. In terms of these matrices, the model implies $Y = UC$ and $YW = U$.

C.2. Simulation of the posterior with the Gibbs sampler

Initialize the simulation with $W^0 = I_N$. Then, in step z

a. draw q_{ns}^z from the conditional posterior $p(q_{ns}|Y, W^{z-1})$, for $n = 1, \ldots, N$, $s = 1, \ldots, S$, collect them in matrix Q^z;
b. draw W^z from the conditional posterior $p(W|Y, Q^z)$;
c. compute the objects of interest $C^z = (W^z)^{-1}$ and $U^z = YW^z$.

C.3. The conditional posterior of W

The conditional posterior of W is a nonstandard density

$$p(W|Y, Q_1, \ldots, Q_T) \propto |W|^T \exp\left(-\frac{1}{2}\sum_{t=1}^{T} y_t' W Q_t W' y_t\right) p(W). \tag{A23}$$

Drawing from this density is the only non-trivial step in the simulation. Following Jarociński (2024) we draw a candidate W^* from the Gaussian proposal density

$$f(W) = \mathcal{N}\left(\hat{w}, (-\mathcal{H})^{-1}\right) \tag{A24}$$

where $\hat{w}$ is the mode of $p(W|Y, \cdot)$ and $\mathcal{H}$ the Hessian of $\log p(W|Y, \cdot)$. The key point is that the Jacobian and Hessian of $\log p(W|Y, \cdot)$ are available in closed form. Consequently, the mode is easy to find by numerical optimization and the Hessian is precisely and quickly computed without resorting to numerical derivatives.

The draw is accepted with probability

$$\max\left(1, \frac{p(W^*|Y,\cdot)\, f(W)}{f(W^*)\, p(W|Y,\cdot)}\right). \tag{A25}$$

The literature has taken different approaches to drawing W (often denoted as A_0). Brunnermeier et al. (2021) use a standard Metropolis-Hastings algorithm and the convergence is slow. Lütkepohl and Woźniak (2020) draw a candidate from a Gaussian distribution centered at the previous draw and with variance $\sum_t WQ_tW'$. The appealing feature of the present approach is that it constructs a substantially better proposal distribution (with the correct mean and variance) at a modest computational cost.

Using the methods in Magnus and Neudecker (2019) it is tedious but straightforward to find that the Jacobian of $\log p(W|Y, Q_1, ..., Q_T)$ is

$$\mathcal{J}(\operatorname{vec} W) = T(\operatorname{vec} W^{-1\prime})' - \sum_t \operatorname{vec}(y_t y_t' W Q_t)' - \operatorname{vec}(W - W_0)'\Omega^{-1} \tag{A26}$$

and the Hessian of $\log p(W|Y, Q_1, ..., Q_T)$ is

$$\mathcal{H}(\operatorname{vec} W) = -TK_{NN}(W^{-1\prime} \otimes W^{-1}) - \sum_t (Q_t \otimes y_t y_t') - \Omega^{-1} \tag{A27}$$

where K_{NN} is a commutation matrix of order N (Magnus and Neudecker 2019).

C.4. The conditional posterior of q_{ns}

The conditional posterior of q_{ns} is a gamma density

$$p(q_{ns}|Y, W) = \mathcal{G}\left((\nu + T_s)/2, 2/(\nu + \sum_{t\in s} u_{nt}^2)\right) \tag{A28}$$

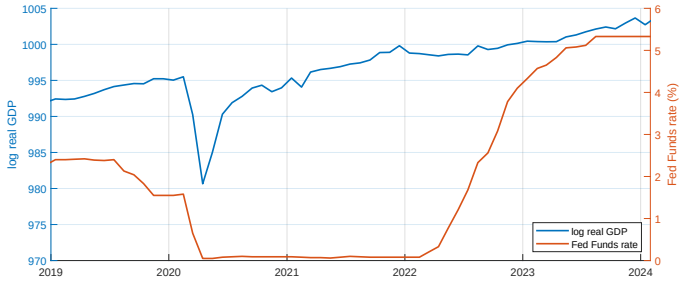
where T_s is the number of observations in this regime, and u_{nt} is the n, t-th element of YW.

Appendix D. Impact of March 2020 surprises on the estimates

In this section, we explain our decision to exclude the March 2020 Fed events from our baseline estimates.

March 2020 features a combination of stronger than expected Fed policy easing and an unprecedented collapse of real activity due to lockdowns. Most importantly, on March 3 the FOMC surprised the markets with a 50 basis points rate cut, against the expectations of a 25 basis points rate cut. Figure A2 below demonstrates the unusual size of the interest rate and real activity fluctuation in this episode. VARs and local projections that include March 2020 find a temporary expansionary effect of contractionary monetary policy shocks, which goes away after excluding March 2020, so we decided to always exclude March 2020 to avoid this outsized impact of a single

FIGURE A2. Real GDP and the federal funds rate, 2019-2024



episode.

Appendix E. Time-Varying Policy Rule

In the baseline framework we treat the Fed's response coefficient α^F as fixed but imperfectly perceived by the market. One might object that rational markets should eventually learn the true coefficient, eliminating systematic misperception. We address this concern by extending the model to allow the rule coefficient to follow a random walk. We show that the CBI results are unchanged, while the FRN results are modified in important ways: predictability of the MPS and attenuation bias in local projections both survive only to the extent that *average* misperception is non-zero—a weaker condition that relies on persistent, rather than permanent, underestimation of the Fed's reaction coefficient. The high-frequency identification strategy remains valid throughout.

Setup. We replace the fixed coefficient α^F with a time-varying coefficient

$$\alpha_t^F = \alpha_{t-1}^F + u_t, \tag{A29}$$

where u_t is i.i.d. with mean zero and variance σ_u^2, and is independent of all other shocks $(\varepsilon_t, \eta_t, \eta_t^F)$. The policy rule is

$$i_t = \alpha_t^F x_t^F + \varepsilon_t. \tag{A30}$$

The central bank announces α_t^F (and all other current shocks) contemporaneously with the policy announcement. Since u_t is not observable to markets before the announcement, their best estimate of the current rule coefficient, formed just before the announcement, is

$$\alpha_{t-\delta}^M = \alpha_{t-1}^F, \tag{A31}$$

so the realized misperception at any given announcement is entirely driven by the current innovation: $\alpha_t^F - \alpha_{t-\delta}^M = u_t$.

MPS decomposition. Subtracting market expectations from the policy rule as in equation (3) yields

$$\underbrace{i_t - E_{t-\delta}^M(i_t)}_{MPS_t} = \underbrace{\varepsilon_t}_{\text{MP shock}} + \underbrace{\alpha_t^F \eta_t^F}_{\text{CBI}} + \underbrace{u_t \cdot x_{t-\delta}^M}_{\text{FRN}}, \tag{A32}$$

where $x_{t-\delta}^M = \rho x_{t-1} - \theta i_{t-1} + \eta_t$ is the market's assessment of the current state. The structure of the three-way decomposition is unchanged, but the FRN term now reflects only the random innovation u_t rather than a fixed wedge $\alpha^F - \alpha^M$.

Predictability. A key implication of the baseline model is that the MPS is predictable using publicly available economic indicators, because the fixed wedge $\alpha^F - \alpha^M > 0$ implies $E[MPS_t \mid x_{t-1}, i_{t-1}, \eta_t] = (\alpha^F - \alpha^M) x_{t-\delta}^M > 0$ when economic conditions are

favorable. Under the random walk extension this calculation gives instead

$$E[MPS_t \mid x_{t-1}, i_{t-1}, \eta_t] = E[u_t] \cdot x^M_{t-\delta} = 0, \tag{A33}$$

since u_t has mean zero and is independent of all public information available before the announcement. *Predictability therefore vanishes entirely when average misperception is zero.* It survives only if there is a non-zero *average* wedge $E[\alpha^F_t - \alpha^M_{t-\delta}] > 0$ persisting throughout the sample, while period-by-period variation in the misperception (u_t) is irrelevant for predictability.

This distinction has direct empirical relevance. Bauer and Swanson (2023a,b) document significant predictability of high-frequency monetary policy surprises in the post-1994 FOMC subsample, which they interpret as evidence of persistent market underestimation of the Fed's reaction coefficient over that period. Our framework is consistent with this interpretation: the predictability they find reflects a positive *average* misperception during their sample, not a fixed structural feature of the economy. Indeed, predictability is notably weaker in the full event sample that extends beyond their subsample, suggesting that average misperception may be smaller or closer to zero over a longer horizon. This observation counsels against treating FRN-induced predictability as a permanent feature of monetary policy surprises.

Attenuation bias. To assess the bias in local projections, we derive the forecast revision of the future state. Replacing the fixed α^F with α^F_t and noting that $\alpha^M_{t-\delta} = \alpha^F_{t-1}$, the forecast revision orthogonalized with respect to the controls (x_{t-1}, i_{t-1}) becomes (compare equation (6))

$$\overline{FR}_t = -\theta\varepsilon_t + (\rho - \theta\alpha^F_t)\eta^F_t + (\rho - \theta\alpha^F_t)\eta_t, \tag{A34}$$

while the orthogonalized MPS is

$$\overline{MPS}_t = \varepsilon_t + \alpha^F_t\eta^F_t + u_t\eta_t. \tag{A35}$$

Note that the component $u_t(\rho x_{t-1} - \theta i_{t-1})$ is absorbed by the controls, leaving only $u_t\eta_t$ in the orthogonalized FRN term. The population LP coefficient is then

$$\begin{aligned} E(\theta^{LP}) &= \frac{\mathrm{cov}(\overline{FR}_t, \overline{MPS}_t)}{\mathrm{var}(\overline{MPS}_t)} \\ &= \frac{-\theta\sigma^2_\varepsilon + E\big[(\rho - \theta\alpha^F_t)\alpha^F_t\big]\,\sigma^2_{\eta^F} + E\big[(\rho - \theta\alpha^F_t)u_t\big]\,\sigma^2_\eta}{\sigma^2_\varepsilon + E\big[(\alpha^F_t)^2\big]\,\sigma^2_{\eta^F} + \sigma^2_u\sigma^2_\eta}. \end{aligned} \tag{A36}$$

The three terms in the numerator correspond to the three shocks. For the FRN term, since $\alpha^F_t = \alpha^F_{t-1} + u_t$ and α^F_{t-1} is independent of u_t:

$$E\Big[(\rho - \theta\alpha^F_t)u_t\Big] = \rho\,E[u_t] - \theta\,E[\alpha^F_{t-1}u_t] - \theta\,E[u^2_t] = -\theta\sigma^2_u. \tag{A37}$$

Substituting back, the FRN contribution to the numerator is $-\theta\sigma_u^2\sigma_\eta^2$ and to the denominator is $\sigma_u^2\sigma_\eta^2$. These exactly cancel in the ratio, contributing a term of $-\theta$: the FRN component of the random walk model yields an unbiased contribution to the LP estimate. Intuitively, the innovation u_t is equally likely to make policy tighter or looser than expected, so it acts as an additional source of exogenous variation rather than a confound. The only remaining source of upward bias is the CBI term:

$$E(\theta^{LP}) = -\theta + \frac{E\big[(\rho - \theta\alpha_t^F)\alpha_t^F\big]\,\sigma_{\eta^F}^2}{\mathrm{var}(\overline{MPS}_t)} > -\theta. \tag{A38}$$

Attenuation bias therefore survives the random walk extension, but it originates exclusively from the CBI channel. *FRN-induced bias requires the same condition as predictability: a non-zero average misperception $E[\alpha_t^F - \alpha_{t-\delta}^M] > 0$. To the extent that average misperception diminishes over the full sample, so does the FRN contribution to the bias.* This reinforces the empirical importance of separately identifying and controlling for the CBI effect, which remains present regardless of whether average misperception is zero.

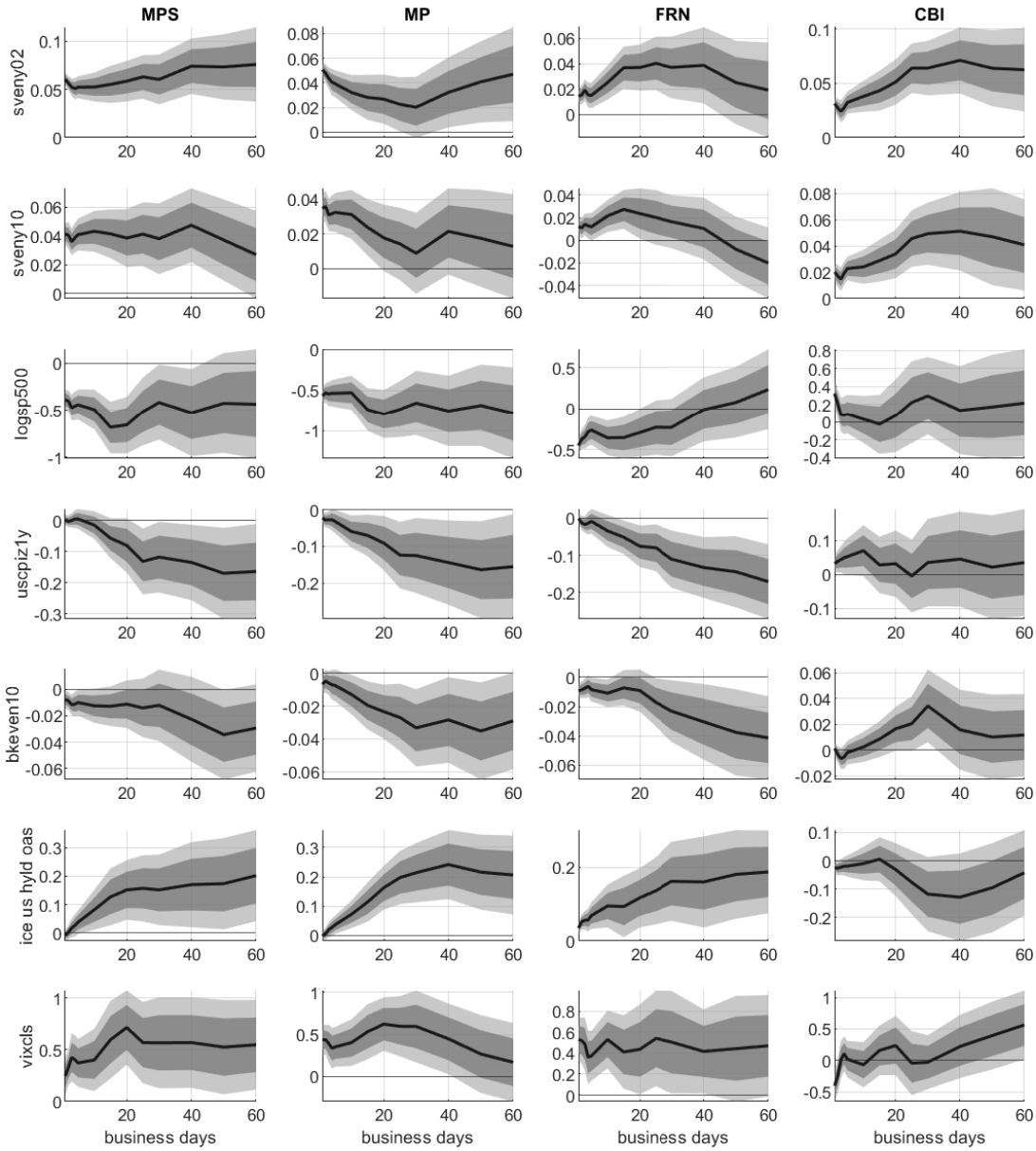
High-frequency identification. The high-frequency stock price decomposition (equation (8)) under the random walk becomes

$$s_t - s_{t-\delta} = \underbrace{-(\theta+1)\varepsilon_t}_{\text{MP}} + \underbrace{(\rho - (\theta+1)\alpha_t^F)\eta_t^F}_{\text{CBI}} + \underbrace{-\theta\, u_t \cdot x_{t-\delta}^M}_{\text{FRN}}. \tag{A39}$$

The structure is qualitatively unchanged. CBI shocks ($\eta_t^F > 0$) continue to appreciate stock prices under the same reasonable parametrization $\rho > (\theta+1)\alpha_t^F$, while the FRN term depreciates stock prices whenever $u_t > 0$ and economic conditions are favorable ($x_{t-\delta}^M > 0$). Crucially, the absence of a direct positive public news effect in the announcement window continues to hold: since η_t is incorporated into prices before the announcement, only the surprise component $u_t\eta_t$ enters the high-frequency window, and this moves rates and stock prices in opposite directions. Our identification strategy, which exploits the sign difference between CBI and FRN shocks in the high-frequency window, therefore remains valid under the random walk extension.

Appendix F. Additional results

FIGURE A3. Daily local projections, $y_{t+h} - y_{t-1} = \alpha + \beta_h u_t + \varepsilon_t^h$: Median target shocks satisfying sign and magnitude restrictions of section 4.1. Baseline sample.



Point estimates of the response (β_h), with heteroskedasticity robust 68% and 90% bands. Responses to raw interest rate surprises (MPS), monetary policy (MP) shocks, Fed response to news (FRN) shocks, and central bank information (CBI) shocks.

FIGURE A4. Dynamic effects of MP, FRN and CBI shocks in a monthly VAR. FOMC sample.

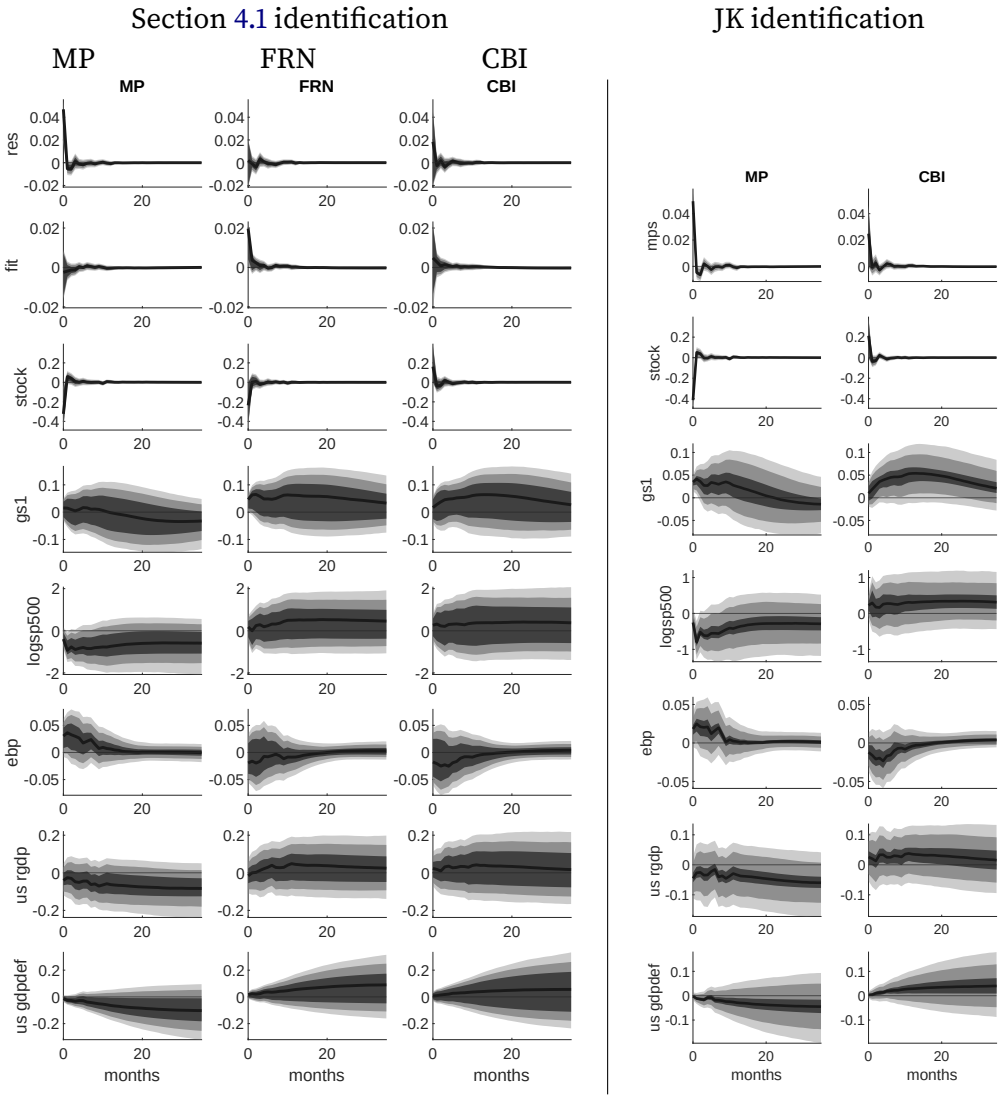
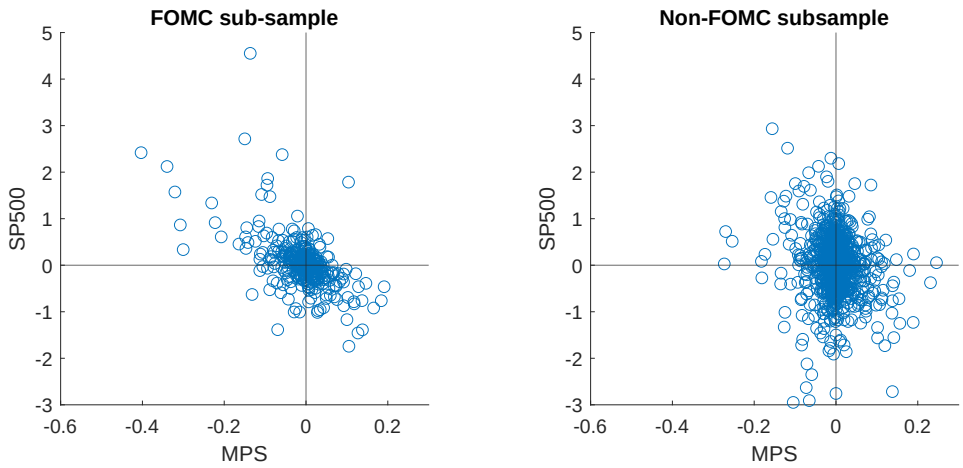



FIGURE A5. Scatter plots of interest rate (MPS) and stock price surprises (SP500) in FOMC and non-FOMC subsamples (all non-FOMC events)